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Satoru Watanabe

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14. Detecting and Estimating Signals over Noisy and Unreliable Synapses: Information-Theoretic Analysis

Amit Manwani, Christof Koch

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The temporal precision with which neurons respond to synaptic inputs has a direct bearing on the nature of the neural code. A characterization of the neuronal noise sources associated with different sub-cellular components (synapse, dendrite, soma, axon, and so on) is needed to understand the relationship between noise and information transfer. Here we study the effect of the unreliable, probabilistic nature of synaptic transmission on information transfer in the absence of interaction among presynaptic inputs. We derive theoretical lower bounds on the capacity of a simple model of a cortical synapse under two different paradigms. In signal estimation, the signal is assumed to be encoded in the mean firing rate of the presynaptic neuron, and the objective is to estimate the continuous input signal from the postsynaptic voltage. In signal detection, the input is binary, and the presence or absence of a presynaptic action potential is to be detected from the postsynaptic voltage. The efficacy of information transfer in synaptic transmission is characterized by deriving optimal strategies under these two paradigms. On the basis of parameter values derived from neocortex, we find that single cortical synapses cannot transmit information reliably, but redundancy obtained using a small number of multiple synapses leads to a significant improvement in the information capacity of synaptic transmission.

15. An Algorithm for Modifying Neurotransmitter Release Probability Based on Pre- and Postsynaptic Spike Timing

Walter Senn, Henry Markram, Misha Tsodyks

https://doi.org/10.1162/089976601300014628 | pp. 35-67

The precise times of occurrence of individual pre- and postsynaptic action potentials are known to play a key role in the modification of synaptic efficacy. Based on stimulation protocols of two synaptically connected neurons, we infer an algorithm that reproduces the experimental data by modifying the probability of vesicle discharge as a function of the relative timing of spikes in the pre- and postsynaptic neurons. The primary feature of this algorithm is an asymmetry with respect to the direction of synaptic modification depending on whether the presynaptic spikes precede or follow the postsynaptic spike. Specifically, if the presynaptic spike occurs up to 50 ms before the postsynaptic spike, the probability of vesicle discharge is upregulated, while the probability of vesicle discharge is downregulated if the presynaptic spike occurs up to 50 ms after the postsynaptic spike. When neurons fire irregularly with Poisson spike trains at constant mean firing rates, the probability of vesicle discharge converges toward a characteristic value determined by the pre- and postsynaptic firing rates. On the other hand, if the mean rates of the Poisson spike trains slowly change with time, our algorithm predicts modifications in the probability of release that generalize Hebbian and Bienenstock-Cooper-Munro rules. We conclude that the proposed spike-based synaptic learning algorithm provides a general framework for regulating neurotransmitter release probability.

16. Differential Filtering of Two Presynaptic Depression Mechanisms

Richard Bertram

https://doi.org/10.1162/089976601300014637 | pp. 69-85

The filtering of input signals carried out at synapses is key to the information processing performed by networks of neurons. Two forms of presynaptic depression, vesicle depletion and G-protein inhibition of Ca2+ channels, can play important roles in the presynaptic processing of information. Using computational models, we demonstrate that these two forms of depression filter information in very different ways. G-protein inhibition acts as a high-pass filter, preferentially transmitting high-frequency input signals to the postsynaptic cell, while vesicle depletion acts as a low-pass filter. We examine how these forms of depression separately and together affect the steady-state postsynaptic responses to trains of stimuli over a range of frequencies. Finally, we demonstrate how differential filtering permits the multiplexing of information within a single impulse train.

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17. A Statistical Theory of Long-Term Potentiation and Depression

John M. Beggs

https://doi.org/10.1162/089976601300014646 | pp. 87-111

The synaptic phenomena of long-term potentiation (LTP) and long-term depression (LTD) have been intensively studied for over twenty-five years. Although many diverse aspects of these forms of plasticity have been observed, no single theory has offered a unifying explanation for them. Here, a statistical "bin" model is proposed to account for a variety of features observed in LTP and LTD experiments performed with field potentials in mammalian cortical slices. It is hypothesized that long-term synaptic changes will be induced when statistically unlikely conjunctions of pre- and postsynaptic activity occur. This hypothesis implies that finite changes in synaptic strength will be proportional to information transmitted by conjunctions and that excitatory synapses will obey a Hebbian rule (Hebb, 1949). Using only one set of constants, the bin model offers an explanation as to why synaptic strength decreases in a decelerating manner during LTD induction (Mulkey & Malenka, 1992); why the induction protocols for LTP and LTD are asymmetric (Dudek & Bear, 1992; Mulkey & Malenka, 1992); why stimulation over a range of frequencies produces a frequency-response curve similar to that proposed by the BCM theory (Bienenstock, Cooper, & Munro, 1982; Dudek & Bear, 1992); and why this curve would shift as postsynaptic activity is changed (Kirkwood, Rioult, & Bear, 1996). In addition, the bin model offers an alternative to the BCM theory by predicting that changes in postsynaptic activity will produce vertical shifts in the curve rather than merely horizontal shifts.

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18. Minimal Model for Intracellular Calcium Oscillations and Electrical Bursting in Melanotrope Cells of Xenopus Laevis

L. Niels Cornelisse, Wim J.J.M. Scheenen, Werner J.H. Koopman, Eric W. Roubos, Stan Gielen

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A minimal model is presented to explain changes in frequency, shape, and amplitude of Ca2+ oscillations in the neuroendocrine melanotrope cell of Xenopus Laevis. It describes the cell as a plasma membrane oscillator with influx of extracellular Ca2+ via voltage-gated Ca2+ channels in the plasma membrane. The Ca2+ oscillations in the Xenopus melanotrope show specific features that cannot be explained by previous

models for electrically bursting cells using one set of parameters. The model assumes a KCa-channel with slow Ca2+-dependent gating kinetics that initiates and terminates the bursts. The slow kinetics of this channel cause an activation of the Kca-channel with a phase shift relative to the intracellular Ca2+ concentration. The phase shift, together with the presence of a Na+ channel that has a lower threshold than the Ca2+ channel, generate the characteristic features of the Ca2+ oscillations in the Xenopus melanotrope cell.

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19. Neural Field Model of Receptive Field Restructuring in Primary Visual Cortex

Katrin Suder, Florentin Wörgötter, Thomas Wennekers

https://doi.org/10.1162/089976601300014664 | pp. 139-159

Receptive fields (RF) in the visual cortex can change their size depending on the state of the individual. This reflects a changing visual resolution according to different demands on information processing during drowsiness. So far, however, the possible mechanisms that underlie these size changes have not been tested rigorously. Only qualitatively has it been suggested that state-dependent lateral geniculate nucleus (LGN) firing patterns (burst versus tonic firing) are mainly responsible for the observed cortical receptive field restructuring. Here, we employ a neural field approach to describe the changes of cortical RF properties analytically. Expressions to describe the spatiotemporal receptive fields are given for pure feedforward networks. The model predicts that visual latencies increase nonlinearly with the distance of the stimulus location from the RF center. RF restructuring effects are faithfully reproduced. Despite the changing RF sizes, the model demonstrates that the width of the spatial membrane potential profile (as measured by the variance sigma of a gaussian) remains constant in cortex. In contrast, it is shown for recurrent networks that both the RF width and the width of the membrane potential profile generically depend on time and can even increase if lateral cortical excitatory connections extend further than fibers from LGN to cortex. In order to differentiate between a feedforward and a recurrent mechanism causing the experimental RF changes, we fitted the data to the analytically derived point-spread functions. Results of the fits provide estimates for model parameters consistent with the literature data and support the hypothesis that the observed RF sharpening is indeed mainly driven by input from LGN, not by recurrent intracortical connections.

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20. On the Phase-Space Dynamics of Systems of Spiking Neurons. I: Model and Experiments

Arunava Banerjee

https://doi.org/10.1162/089976601300014673 | pp. 161-193

We investigate the phase-space dynamics of a general model of local systems of biological neurons in order to deduce the salient dynamical characteristics of such systems. In this article, we present a detailed exposition of an abstract dynamical system that models systems of biological neurons. The abstract system is based on a limited set of realistic assumptions and thus accommodates a wide range of neuronal models. Simulation results are presented for several instantiations of the abstract system, each modeling a typical neocortical column to a different degree of accuracy. The results demonstrate that the dynamics of the systems are generally consistent with that observed in neurophysiological experiments. They reveal that the qualitative behavior of the class of systems can be classified into three distinct categories: quiescence,

intense periodic activity resembling a state of seizure, and sustained chaos over the range of intrinsic activity typically associated with normal operational conditions in the neocortex. We discuss basic ramifications of this result with regard to the computational nature of neocortical neuronal systems.

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21. On the Phase-Space Dynamics of Systems of Spiking Neurons. II: Formal Analysis

Arunava Banerjee

https://doi.org/10.1162/089976601300014682 | pp. 195-225

We begin with a brief review of the abstract dynamical system that models systems of biological neurons, introduced in the original article. We then analyze the dynamics of the system. Formal analysis of local properties of flows reveals contraction, expansion, and folding in different sections of the phase-space. The criterion for the system, set up to model a typical neocortical column, to be sensitive to initial conditions is identified. Based on physiological parameters, we then deduce that periodic orbits in the region of the phase-space corresponding to normal operational conditions in the neocortex are almost surely (with probability 1) unstable, those in the region corresponding to seizure-like conditions are almost surely stable, and trajectories in the region corresponding to normal operational conditions are almost surely sensitive to initial conditions. Next, we present a procedure that isolates all basic sets, complex sets, and attractors incrementally. Based on the two sets of results, we conclude that chaotic attractors that are potentially anisotropic play a central role in the dynamics of such systems. Finally, we examine the impact of this result on the computational nature of neocortical neuronal systems.

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22. Subtractive and Divisive Inhibition: Effect of Voltage-Dependent Inhibitory Conductances and Noise

Brent Doiron, André Longtin, Neil J. Berman, Leonard Maler

https://doi.org/10.1162/089976601300014691 | pp. 227-248

The influence of voltage-dependent inhibitory conductances on firing rate versus input current (f-I) curves is studied using simulations from a new compartmental model of a pyramidal cell of the weakly electric fish Apteronotus leptorhynchus. The voltage dependence of shunting-type inhibition enhances the subtractive effect of inhibition on f-I curves previously demonstrated in Holt and Koch (1997) for the voltage-independent case. This increased effectiveness is explained using the behavior of the average subthreshold voltage with input current and, in particular, the nonlinearity of Ohm's law in the subthreshold regime. Our simulations also reveal, for both voltage-dependent and -independent inhibitory conductances, a divisive inhibition regime at low frequencies ($f < 40$ Hz). This regime, dependent on stochastic inhibitory synaptic input and a coupling of inhibitory strength and variance, gives way to subtractive inhibition at higher-output frequencies ($f > 40$ Hz). A simple leaky integrate-and-fire type model that incorporates the voltage dependence supports the results from our full ionic simulations.

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23. Spatiotemporal Connectionist Networks: A Taxonomy and Review

Stefan C. Kremer

https://doi.org/10.1162/089976601300014538 | pp. 249-306

This article reviews connectionist network architectures and training algorithms that are capable of dealing with patterns distributed across both space and time—spatiotemporal patterns. It provides common mathematical, algorithmic, and illustrative frameworks for describing spatiotemporal networks, making it easier to compare and contrast their representational and operational characteristics. Computational power, representational issues, and learning are discussed. In additional references to the relevant source publications are provided. This article can serve as a guide to prospective users of spatiotemporal networks by providing an overview of the operational and representational alternatives available.

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24. Formulations of Support Vector Machines: A Note from an Optimization Point of View

Chih■Jen Lin

https://doi.org/10.1162/089976601300014547 | pp. 307-317

In this article, we discuss issues about formulations of support vector machines (SVM) from an optimization point of view. First, SVMs map training data into a higher- (maybe infinite-) dimensional space. Currently primal and dual formulations of SVM are derived in the finite dimensional space and readily extend to the infinite-dimensional space. We rigorously discuss the primal-dual relation in the infinite-dimensional spaces. Second, SVM formulations contain penalty terms, which are different from unconstrained penalty functions in optimization. Traditionally unconstrained penalty functions approximate a constrained problem as the penalty parameter increases. We are interested in similar properties for SVM formulations. For two of the most popular SVM formulations, we show that one enjoys properties of exact penalty functions, but the other is only like traditional penalty functions, which converge when the penalty parameter goes to infinity.

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25. Minimal Feedforward Parity Networks Using Threshold Gates

Hon-Kwok Fung, Leong Kwan Li

https://doi.org/10.1162/089976601300014556 | pp. 319-326

This article presents preliminary research on the general problem of reducing the number of neurons needed in a neural network so that the network can perform a specific recognition task. We consider a single-hidden-layer feedforward network in which only McCulloch-Pitts units are employed in the hidden layer. We show that if only interconnections between adjacent layers are allowed, the minimum size of the hidden layer required to solve the n-bit parity problem is n when $n \leq 4$.

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26. Does Corticothalamic Feedback Control Cortical Velocity Tuning?

Ulrich Hillenbrand, J. Leo van Hemmen

https://doi.org/10.1162/089976601300014565 | pp. 327-355

The thalamus is the major gate to the cortex, and its contribution to cortical receptive field properties is well established. Cortical feedback to the thalamus is, in turn, the anatomically dominant input to relay cells, yet its influence on thalamic processing has been difficult to interpret. For an understanding of complex sensory processing, detailed concepts of the corticothalamic interplay need to be established. To study

corticogeniculate processing in a model, we draw on various physiological and anatomical data concerning the intrinsic dynamics of geniculate relay neurons, the cortical influence on relay modes, lagged and nonlagged neurons, and the structure of visual cortical receptive fields. In extensive computer simulations, we elaborate the novel hypothesis that the visual cortex controls via feedback the temporal response properties of geniculate relay cells in a way that alters the tuning of cortical cells for speed.

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27. A Competitive-Layer Model for Feature Binding and Sensory Segmentation

Heiko Wersing, Jochen J. Steil, Helge Ritter

https://doi.org/10.1162/089976601300014574 | pp. 357-387

We present a recurrent neural network for feature binding and sensory segmentation: the competitive-layer model (CLM). The CLM uses topographically structured competitive and cooperative interactions in a layered network to partition a set of input features into salient groups. The dynamics is formulated within a standard additive recurrent network with linear threshold neurons. Contextual relations among features are coded by pairwise compatibilities, which define an energy function to be minimized by the neural dynamics. Due to the usage of dynamical winner-take-all circuits, the model gains more flexible response properties than spin models of segmentation by exploiting amplitude information in the grouping process. We prove analytic results on the convergence and stable attractors of the CLM, which generalize earlier results on winner-take-all networks, and incorporate deterministic annealing for robustness against local minima. The piecewise linear dynamics of the CLM allows a linear eigensubspace analysis, which we use to analyze the dynamics of binding in conjunction with annealing. For the example of contour detection, we show how the CLM can integrate figure-ground segmentation and grouping into a unified model.

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28. Learning Object Representations Using A Priori Constraints Within ORASSYLL

Norbert Krüger

https://doi.org/10.1162/089976601300014583 | pp. 389-410

In this article, a biologically plausible and efficient object recognition system (called ORASSYLL) is introduced, based on a set of a priori constraints motivated by findings of developmental psychology and neurophysiology. These constraints are concerned with the organization of the input in local and corresponding entities, the interpretation of the input by its transformation in a highly structured feature space, and the evaluation of features extracted from an image sequence by statistical evaluation criteria. In the context of the bias-variance dilemma, the functional role of a priori knowledge within ORASSYLL is discussed. In contrast to systems in which object representations are defined manually,the introduced constraints allow an autonomous learning from complex scenes.

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29. Binding and Normalization of Binary Sparse Distributed Representations by Context-Dependent Thinning

Dmitri A. Rachkovskij, Ernst Kussul

https://doi.org/10.1162/089976601300014592 | pp. 411-452

Distributed representations were often criticized as inappropriate for encoding of data with a complex structure. However Plate's holographic reduced representations and Kanerva's binary spatter codes are recent schemes that allow on-the-fly encoding of nested compositional structures by real-valued or dense binary vectors of fixed dimensionality. In this article we consider procedures of the context-dependent thinning developed for representation of complex hierarchical items in the architecture of associative-projective neural networks. These procedures provide binding of items represented by sparse binary codevectors (with low probability of 1s). Such an encoding is biologically plausible and allows a high storage capacity of distributed associative memory where the codevectors may be stored. In contrast to known binding procedures, context-dependent thinning preserves the same low density (or sparseness) of the bound codevector for a varied number of component codevectors. Besides, a bound codevector is similar not only to another one with similar component codevectors (as in other schemes) but also to the component codevectors themselves. This allows the similarity of structures to be estimated by the overlap of their codevectors, without retrieval of the component codevectors. This also allows easy retrieval of the component codevectors. Examples of algorithmic and neural network implementations of the thinning procedures are considered. We also present representation examples for various types of nested structured data (propositions using role filler and predicate arguments schemes, trees, and directed acyclic graphs) using sparse codevectors of fixed dimension. Such representations may provide a fruitful alternative to the symbolic representations of traditional artificial intelligence as well as to the localist and microfeature-based connectionist representations.

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30. Resolution-Based Complexity Control for Gaussian Mixture Models

Peter Meinicke, Helge Ritter

https://doi.org/10.1162/089976601300014600 | pp. 453-475

In the domain of unsupervised learning, mixtures of gaussians have become a popular tool for statistical modeling. For this class of generative models, we present a complexity control scheme, which provides an effective means for avoiding the problem of overfitting usually encountered with unconstrained (mixtures of) gaussians in high dimensions. According to some prespecified level of resolution as implied by a fixed variance noise model, the scheme provides an automatic selection of the dimensionalities of some local signal subspaces by maximum likelihood estimation. Together with a resolution-based control scheme for adjusting the number of mixture components, we arrive at an incremental model refinement procedure within a common deterministic annealing framework, which enables an efficient exploration of the model space. The advantages of the resolution-based framework are illustrated by experimental results on synthetic and high-dimensional real-world data.

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31. The Limits of Counting Accuracy in Distributed Neural Representations

A. R. Gardner■Medwin, H. B. Barlow

https://doi.org/10.1162/089976601300014420 | pp. 477-504

Learning about a causal or statistical association depends on comparing frequencies of joint occurrence with frequencies expected from separate occurrences, and to do this, events must somehow be counted. Physiological mechanisms can easily generate the necessary measures if there is a direct, one-to-one

relationship between significant events and neural activity, but if the events are represented across cell populations in a distributed manner, the counting of one event will be interfered with by the occurrence of others. Although the mean interference can be allowed for, there is inevitably an increase in the variance of frequency estimates that results in the need for extra data to achieve reliable learning. This lowering of statistical efficiency (Fisher, 1925) is calculated as the ratio of the minimum to actual variance of the estimates. We define two neural models, based on presynaptic and Hebbian synaptic modification, and explore the effects of sparse coding and the relative frequencies of events on the efficiency of frequency estimates. High counting efficiency must be a desirable feature of biological representations, but the results show that the number of events that can be counted simultaneously with 50% efficiency is fewer than the number of cells or 0.1-0.25 of the number of synapses (on the two models) - many fewer than can be unambiguously represented. Direct representations would lead to greater counting efficiency, but distributed representations have the versatility of detecting and counting many unforeseen or rare events. Efficient counting of rare but important events requires that they engage more active cells than common or unimportant ones. The results suggest reasons that representations in the cerebral cortex appear to use extravagant numbers of cells and modular organization, and they emphasize the importance of neuronal trigger features and the phenomena of habituation and attention.

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32. An Expectation-Maximization Approach to Nonlinear Component Analysis

Roman Rosipal, Mark Girolami

https://doi.org/10.1162/089976601300014439 | pp. 505-510

The proposal of considering nonlinear principal component analysis as a kernel eigenvalue problem has provided an extremely powerful method of extracting nonlinear features for a number of classification and regression applications. Whereas the utilization of Mercer kernels makes the problem of computing principal components in, possibly, infinite-dimensional feature spaces tractable, there are still the attendant numerical problems of diagonalizing large matrices. In this contribution, we propose an expectation-maximization approach for performing kernel principal component analysis and show this to be a computationally efficient method, especially when the number of data points is large.

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33. A Population Density Approach That Facilitates Large-Scale Modeling of Neural Networks: Extension to Slow Inhibitory Synapses

Duane Q. Nykamp, Daniel Tranchina

https://doi.org/10.1162/089976601300014448 | pp. 511-546

A previously developed method for efficiently simulating complex networks of integrate-and-fire neurons was specialized to the case in which the neurons have fast unitary postsynaptic conductances. However, inhibitory synaptic conductances are often slower than excitatory ones for cortical neurons, and this difference can have a profound effect on network dynamics that cannot be captured with neurons that have only fast synapses. We thus extend the model to include slow inhibitory synapses. In this model, neurons are grouped into large populations of similar neurons. For each population, we calculate the evolution of a probability density function (PDF), which describes the distribution of neurons over state-space. The population firing rate is given by the flux of probability across the threshold voltage for firing an action

potential. In the case of fast synaptic conductances, the PDF was one-dimensional, as the state of a neuron was completely determined by its transmembrane voltage. An exact extension to slow inhibitory synapses increases the dimension of the PDF to two or three, as the state of a neuron now includes the state of its inhibitory synaptic conductance. However, by assuming that the expected value of a neuron's inhibitory conductance is independent of its voltage, we derive a reduction to a one-dimensional PDF and avoid increasing the computational complexity of the problem. We demonstrate that although this assumption is not strictly valid, the results of the reduced model are surprisingly accurate.

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34. A Complex Cell-Like Receptive Field Obtained by Information Maximization

Kenji Okajima, Hitoshi Imaoka

https://doi.org/10.1162/089976601300014457 | pp. 547-562

The energy model (Pollen & Ronner, 1983; Adelson & Bergen, 1985) for a complex cell in the visual cortex is investigated theoretically. The energy model describes the output of a complex cell as the squared sum of outputs of two linear operators. An information-maximization problem to determine the two linear operators is investigated assuming the low signal-to-noise ratio limit and a localization term in the objective function. As a result, two linear operators characterized by a quadrature pair of Gabor functions are obtained as solutions. The result agrees with the energy model, which well describes the shift-invariant and orientation-selective responses of actual complex cells, and thus suggests that complex cells are optimally designed from an information-theoretic viewpoint.

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35. Self-Organization of Topographic Mixture Networks Using Attentional Feedback

James R. Williamson

https://doi.org/10.1162/089976601300014466 | pp. 563-593

This article proposes a neural network model of supervised learning that employs biologically motivated constraints of using local, on-line, constructive learning. The model possesses two novel learning mechanisms. The first is a network for learning topographic mixtures. The network's internal category nodes are the mixture components, which learn to encode smooth distributions in the input space by taking advantage of topography in the input feature maps. The second mechanism is an attentional biasing feedback circuit. When the network makes an incorrect output prediction, this feedback circuit modulates the learning rates of the category nodes, by amounts based on the sharpness of their tuning, in order to improve the network's prediction accuracy. The network is evaluated on several standard classification benchmarks and shown to perform well in comparison to other classifiers.

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36. Auto-SOM: Recursive Parameter Estimation for Guidance of Self-Organizing Feature Maps

Karin Haese, Geoffrey J. Goodhill

https://doi.org/10.1162/089976601300014475 | pp. 595-619

An important technique for exploratory data analysis is to form a mapping from the high-dimensional data space to a low-dimensional representation space such that neighborhoods are preserved. A popular method for achieving this is Kohonen's self-organizing map (SOM) algorithm. However, in its original form, this requires the user to choose the values of several parameters heuristically to achieve good performance. Here we present the Auto-SOM, an algorithm that estimates the learning parameters during the training of SOMs automatically. The application of Auto-SOM provides the facility to avoid neighborhood violations up to a user-defined degree in either mapping direction. Auto-SOM consists of a Kalman filter implementation of the SOM coupled with a recursive parameter estimation method. The Kalman filter trains the neurons' weights with estimated learning coefficients so as to minimize the variance of the estimation error. The recursive parameter estimation method estimates the width of the neighborhood function by minimizing the prediction error variance of the Kalman filter. In addition, the "topographic function" is incorporated to measure neighborhood violations and prevent the map's converging to configurations with neighborhood violations. It is demonstrated that neighborhoods can be preserved in both mapping directions as desired for dimension-reducing applications. The development of neighborhood-preserving maps and their convergence behavior is demonstrated by three examples accounting for the basic applications of self-organizing feature maps.

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37. Exponential Convergence of Delayed Dynamical Systems

Tianping Chen, ■■■-■■■ ■■■■■

https://doi.org/10.1162/089976601300014484 | pp. 621-635

We discuss some delayed dynamical systems, investigating their stability and convergence. We prove that under mild conditions, these delayed systems are global exponential convergent.

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38. Improvements to Platt's SMO Algorithm for SVM Classifier Design

S. Sathiya Keerthi, Shirish Shevade, Chiranjib Bhattacharyya, K. R. K. Murthy

https://doi.org/10.1162/089976601300014493 | pp. 637-649

This article points out an important source of inefficiency in Platt's sequential minimal optimization (SMO) algorithm that is caused by the use of a single threshold value. Using clues from the KKT conditions for the dual problem, two threshold parameters are employed to derive modifications of SMO. These modified algorithms perform significantly faster than the original SMO on all benchmark data sets tried.

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39. Learning Hough Transform: A Neural Network Model

Jayanta Kumar Basak

https://doi.org/10.1162/089976601300014501 | pp. 651-676

A single-layered Hough transform network is proposed that accepts image coordinates of each object pixel as input and produces a set of outputs that indicate the belongingness of the pixel to a particular structure (e.g., a straight line). The network is able to learn adaptively the parametric forms of the linear segments present in the image. It is designed for learning and identification not only of linear segments in two-dimensional images but also the planes and hyperplanes in the higher-dimensional spaces. It provides

an efficient representation of visual information embedded in the connection weights. The network not only reduces the large space requirement, as in the case of classical Hough transform, but also represents the parameters with high precision.

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40. A Constrained EM Algorithm for Independent Component Analysis

Max Welling, Markus Weber

https://doi.org/10.1162/089976601300014510 | pp. 677-689

We introduce a novel way of performing independent component analysis using a constrained version of the expectation-maximization (EM) algorithm. The source distributions are modeled as D one-dimensional mixtures of gaussians. The observed data are modeled as linear mixtures of the sources with additive, isotropic noise. This generative model is fit to the data using constrained EM. The simpler "soft-switching" approach is introduced, which uses only one parameter to decide on the sub- or supergaussian nature of the sources. We explain how our approach relates to independent factor analysis.

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41. Emergence of Memory-Driven Command Neurons in Evolved Artificial Agents

Ranit Aharonov-Barki, Tuvik Beker, Eytan Ruppin

https://doi.org/10.1162/089976601300014529 | pp. 691-716

Using evolutionary simulations, we develop autonomous agents controlled by artificial neural networks (ANNs). In simple lifelike tasks of foraging and navigation, high performance levels are attained by agents equipped with fully recurrent ANN controllers. In a set of experiments sharing the same behavioral task but differing in the sensory input available to the agents, we find a common structure of a command neuron switching the dynamics of the network between radically different behavioral modes. When sensory position information is available, the command neuron reflects a map of the environment, acting as a location-dependent cell sensitive to the location and orientation of the agent. When such information is unavailable, the command neuron's activity is based on a spontaneously evolving short-term memory mechanism, which underlies its apparent place-sensitive activity. A two-parameter stochastic model for this memory mechanism is proposed. We show that the parameter values emerging from the evolutionary simulations are near optimal; evolution takes advantage of seemingly harmful features of the environment to maximize the agent's foraging efficiency. The accessibility of evolved ANNs for a detailed inspection, together with the resemblance of some of the results to known findings from neurobiology, places evolved ANNs as an excellent candidate model for the study of structure and function relationship in complex nervous systems.

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42. Sampling Properties of the Spectrum and Coherency of Sequences of Action Potentials

M. R. Jarvis, Partha P. Mitra

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The spectrum and coherency are useful quantities for characterizing the temporal correlations and functional relations within and between point processes. This article begins with a review of these quantities, their interpretation, and how they may be estimated. A discussion of how to assess the statistical significance of features in these measures is included. In addition, new work is presented that builds on the framework established in the review section. This work investigates how the estimates and their error bars are modified by finite sample sizes. Finite sample corrections are derived based on a doubly stochastic inhomogeneous Poisson process model in which the rate functions are drawn from a low-variance gaussian process. It is found that in contrast to continuous processes, the variance of the estimators cannot be reduced by smoothing beyond a scale set by the number of point events in the interval. Alternatively, the degrees of freedom of the estimators can be thought of as bounded from above by the expected number of point events in the interval. Further new work describing and illustrating a method for detecting the presence of a line in a point process spectrum is also presented, corresponding to the detection of a periodic modulation of the underlying rate. This work demonstrates that a known statistical test, applicable to continuous processes, applies with little modification to point process spectra and is of utility in studying a point process driven by a continuous stimulus. Although the material discussed is of general applicability to point processes, attention will be confined to sequences of neuronal action potentials (spike trains), the motivation for this work.

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43. A Novel Spike Distance

Mark C. W. van Rossum

https://doi.org/10.1162/089976601300014321 | pp. 751-763

The discrimination between two spike trains is a fundamental problem for both experimentalists and the nervous system itself. We introduce a measure for the distance between two spike trains. The distance has a time constant as a parameter. Depending on this parameter, the distance interpolates between a coincidence detector and a rate difference counter. The dependence of the distance on noise is studied with an integrate-and-fire model. For an intermediate range of the time constants, the distance depends linearly on the noise. This property can be used to determine the intrinsic noise of a neuron.

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44. On Synchrony of Weakly Coupled Neurons at Low Firing Rate

L. Neltner, David Hansel

https://doi.org/10.1162/089976601300014330 | pp. 765-774

The dynamics of a pair of weakly interacting conductance-based neurons, firing at low frequency, nu, is investigated in the framework of the phase-reduction method. The stability of the antiphase and the in-phase locked state is studied. It is found that for a large class of conductance-based models, the antiphase state is stable (resp., unstable) for excitatory (resp., inhibitory) interactions if the synaptic time constant is above a critical value tau(c)(s), which scales as the absolute value of log nu when nu goes to zero.

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45. Population Coding with Correlation and an Unfaithful Model

Si Wu, Hiroyuki Nakahara, ■■■-■■■ ■■■■■

https://doi.org/10.1162/089976601300014349 | pp. 775-797

This study investigates a population decoding paradigm in which the maximum likelihood inference is based on an unfaithful decoding model (UMLI). This is usually the case for neural population decoding because the encoding process of the brain is not exactly known or because a simplified decoding model is preferred for saving computational cost. We consider an unfaithful decoding model that neglects the pair-wise correlation between neuronal activities and prove that UMLI is asymptotically efficient when the neuronal correlation is uniform or of limited range. The performance of UMLI is compared with that of the maximum likelihood inference based on the faithful model and that of the center-of-mass decoding method. It turns out that UMLI has advantages of decreasing the computational complexity remarkably and maintaining high-level decoding accuracy. Moreover, it can be implemented by a biologically feasible recurrent network (Pouget, Zhang, Deneve, & Latham, 1998). The effect of correlation on the decoding accuracy is also discussed.

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46. Metabolically Efficient Information Processing

Vijay Balasubramanian, Don Kimber, Michael J. Berry

https://doi.org/10.1162/089976601300014358 | pp. 799-815

Energy-efficient information transmission may be relevant to biological sensory signal processing as well as to low-power electronic devices. We explore its consequences in two different regimes. In an "immediate" regime, we argue that the information rate should be maximized subject to a power constraint, and in an "exploratory" regime, the transmission rate per power cost should be maximized. In the absence of noise, discrete inputs are optimally encoded into Boltzmann distributed output symbols. In the exploratory regime, the partition function of this distribution is numerically equal to 1. The structure of the optimal code is strongly affected by noise in the transmission channel. The Arimoto-Blahut algorithm, generalized for cost constraints, can be used to derive and interpret the distribution of symbols for optimal energy-efficient coding in the presence of noise. We outline the possibilities and problems in extending our results to information coding and transmission in neurobiological systems.

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47. Effective Neuronal Learning with Ineffective Hebbian Learning Rules

Gal Chechik, Isaac Meilijson, Eytan Ruppin

https://doi.org/10.1162/089976601300014367 | pp. 817-840

In this article we revisit the classical neuroscience paradigm of Hebbian learning. We find that it is difficult to achieve effective associative memory storage by Hebbian synaptic learning, since it requires network-level information at the synaptic level or sparse coding level. Effective learning can yet be achieved even with nonsparse patterns by a neuronal process that maintains a zero sum of the incoming synaptic efficacies. This weight correction improves the memory capacity of associative networks from an essentially bounded one to a memory capacity that scales linearly with network size. It also enables the effective storage of patterns with multiple levels of activity within a single network. Such neuronal weight correction can be successfully carried out by activity-dependent homeostasis of the neuron's synaptic efficacies, which was recently observed in cortical tissue. Thus, our findings suggest that associative learning by Hebbian synaptic learning should be accompanied by continuous remodeling of neuronally driven regulatory processes in the brain.

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48. Temporal Difference Model Reproduces Anticipatory Neural Activity

Roland E. Suri, Wolfram Schultz

https://doi.org/10.1162/089976601300014376 | pp. 841-862

Anticipatory neural activity preceding behaviorally important events has been reported in cortex, striatum, and midbrain dopamine neurons. Whereas dopamine neurons are phasically activated by reward-predictive stimuli, anticipatory activity of cortical and striatal neurons is increased during delay periods before important events. Characteristics of dopamine neuron activity resemble those of the prediction error signal of the temporal difference (TD) model of Pavlovian learning (Sutton & Barto, 1990). This study demonstrates that the prediction signal of the TD model reproduces characteristics of cortical and striatal anticipatory neural activity. This finding suggests that tonic anticipatory activities may reflect prediction signals that are involved in the processing of dopamine neuron activity.

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49. Blind Source Separation by Sparse Decomposition in a Signal Dictionary

Michael Zibulevsky, Barak A. Pearlmutter

https://doi.org/10.1162/089976601300014385 | pp. 863-882

The blind source separation problem is to extract the underlying source signals from a set of linear mixtures, where the mixing matrix is unknown. This situation is common in acoustics, radio, medical signal and image processing, hyperspectral imaging, and other areas. We suggest a two-stage separation process: a priori selection of a possibly overcomplete signal dictionary (for instance, a wavelet frame or a learned dictionary) in which the sources are assumed to be sparsely representable, followed by unmixing the sources by exploiting the their sparse representability. We consider the general case of more sources than mixtures, but also derive a more efficient algorithm in the case of a nonovercomplete dictionary and an equal numbers of sources and mixtures. Experiments with artificial signals and musical sounds demonstrate significantly better separation than other known techniques.

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50. Complexity Pursuit: Separating Interesting Components from Time Series

Aapo Hyvärinen

https://doi.org/10.1162/089976601300014394 | pp. 883-898

A generalization of projection pursuit for time series, that is, signals with time structure, is introduced. The goal is to find projections of time series that have interesting structure, defined using criteria related to Kolmogoroff complexity or coding length. Interesting signals are those that can be coded with a short code length. We derive a simple approximation of coding length that takes into account both the nongaussianity and the autocorrelations of the time series. Also, we derive a simple algorithm for its approximative optimization. The resulting method is closely related to blind separation of nongaussian, time-dependent source signals.

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51. Algebraic Analysis for Nonidentifiable Learning Machines

Sumio Watanabe

https://doi.org/10.1162/089976601300014402 | pp. 899-933

This article clarifies the relation between the learning curve and the algebraic geometrical structure of an unidentifiable learning machine such as a multilayer neural network whose true parameter set is an analytic set with singular points. By using a concept in algebraic analysis, we rigorously prove that the Bayesian stochastic complexity or the free energy is asymptotically equal to lambda(1) log n - (m(1) - 1) log log n + constant, where n is the number of training samples and lambda(1) and m(1) are the rational number and the natural number, which are determined as the birational invariant values of the singularities in the parameter space. Also we show an algorithm to calculate lambda(1) and m(1) based on the resolution of singularities in algebraic geometry. In regular statistical models, 2lambda(1) is equal to the number of parameters and m(1) = 1, whereas in nonregular models, such as multilayer networks, 2lambda(1) is not larger than the number of parameters and m(1) > or = 1. Since the increase of the stochastic complexity is equal to the learning curve or the generalization error, the nonidentifiable learning machines are better models than the regular ones if Bayesian ensemble learning is applied.

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52. A New On-Line Learning Model

Shahar Mendelson

https://doi.org/10.1162/089976601300014411 | pp. 935-957

We introduce a new supervised learning model that is a nonhomogeneous Markov process and investigate its properties. We are interested in conditions that ensure that the process converges to a "correct state," which means that the system agrees with the teacher on every "question." We prove a sufficient condition for almost sure convergence to a correct state and give several applications to the convergence theorem. In particular, we prove several convergence results for well-known learning rules in neural networks.

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53. Patterns of Synchrony in Neural Networks with Spike Adaptation

Carl van Vreeswijk, D. Hansel

https://doi.org/10.1162/08997660151134280 | pp. 959-992

We study the emergence of synchronized burst activity in networks of neurons with spike adaptation. We show that networks of tonically firing adapting excitatory neurons can evolve to a state where the neurons burst in a synchronized manner. The mechanism leading to this burst activity is analyzed in a network of integrate-and-fire neurons with spike adaptation. The dependence of this state on the different network parameters is investigated, and it is shown that this mechanism is robust against inhomogeneities, sparseness of the connectivity, and noise. In networks of two populations, one excitatory and one inhibitory, we show that decreasing the inhibitory feedback can cause the network to switch from a tonically active, asynchronous state to the synchronized bursting state. Finally, we show that the same mechanism also causes synchronized burst activity in networks of more realistic conductance-based model neurons.

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54. Bayesian Analysis of Mixtures of Factor Analyzers

Akio Utsugi, Toru Kumagai

https://doi.org/10.1162/08997660151134299 | pp. 993-1002

For Bayesian inference on the mixture of factor analyzers, natural conjugate priors on the parameters are introduced, and then a Gibbs sampler that generates parameter samples following the posterior is constructed. In addition, a deterministic estimation algorithm is derived by taking modes instead of samples from the conditional posteriors used in the Gibbs sampler. This is regarded as a maximum a posteriori estimation algorithm with hyperparameter search. The behaviors of the Gibbs sampler and the deterministic algorithm are compared on a simulation experiment.

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55. Synchronization in Relaxation Oscillator Networks with Conduction Delays

Jeffrey J. Fox, C. Jayaprakash, DeLiang Wang, Shannon R. Campbell

https://doi.org/10.1162/08997660151134307 | pp. 1003-1021

We study locally coupled networks of relaxation oscillators with excitatory connections and conduction delays and propose a mechanism for achieving zero phase-lag synchrony. Our mechanism is based on the observation that different rates of motion along different nullclines of the system can lead to synchrony in the presence of conduction delays. We analyze the system of two coupled oscillators and derive phase compression rates. This analysis indicates how to choose nullclines for individual relaxation oscillators in order to induce rapid synchrony. The numerical simulations demonstrate that our analytical results extend to locally coupled networks with conduction delays and that these networks can attain rapid synchrony with appropriately chosen nullclines and initial conditions. The robustness of the proposed mechanism is verified with respect to different nullclines, variations in parameter values, and initial conditions.

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56. Predictions of the Spontaneous Symmetry-Breaking Theory for Visual Code Completeness and Spatial Scaling in Single-Cell Learning Rules

C. J. S. Webber

https://doi.org/10.1162/08997660151134316 | pp. 1023-1043

This article shows analytically that single-cell learning rules that give rise to oriented and localized receptive fields, when their synaptic weights are randomly and independently initialized according to a plausible assumption of zero prior information, will generate visual codes that are invariant under two-dimensional translations, rotations, and scale magnifications, provided that the statistics of their training images are sufficiently invariant under these transformations. Such codes span different image locations, orientations, and size scales with equal economy. Thus, single-cell rules could account for the spatial scaling property of the cortical simple-cell code. This prediction is tested computationally by training with natural scenes; it is demonstrated that a single-cell learning rule can give rise to simple-cell receptive fields spanning the full range of orientations, image locations, and spatial frequencies (except at the extreme high and low frequencies at which the scale invariance of the statistics of digitally sampled images must ultimately break down, because of the image boundary and the finite pixel resolution). Thus, no constraint on completeness, or any other coupling between cells, is necessary to induce the visual code to span wide ranges of locations, orientations, and size scales. This prediction is made using the theory of spontaneous symmetry breaking, which we have previously shown can also explain the data-driven self-organization of a wide variety of transformation invariances in neurons' responses, such as the translation invariance of complex cell response.

57. Localist Attractor Networks

Richard S. Zemel, Michael C. Mozer

https://doi.org/10.1162/08997660151134325 | pp. 1045-1064

Attractor networks, which map an input space to a discrete output space, are useful for pattern completion--cleaning up noisy or missing input features. However, designing a net to have a given set of attractors is notoriously tricky; training procedures are CPU intensive and often produce spurious attractors and ill-conditioned attractor basins. These difficulties occur because each connection in the network participates in the encoding of multiple attractors. We describe an alternative formulation of attractor networks in which the encoding of knowledge is local, not distributed. Although localist attractor networks have similar dynamics to their distributed counterparts, they are much easier to work with and interpret. We propose a statistical formulation of localist attractor net dynamics, which yields a convergence proof and a mathematical interpretation of model parameters. We present simulation experiments that explore the behavior of localist attractor networks, showing that they yield few spurious attractors, and they readily exhibit two desirable properties of psychological and neurobiological models: priming (faster convergence to an attractor if the attractor has been recently visited) and gang effects (in which the presence of an attractor enhances the attractor basins of neighboring attractors).

58. Stochastic Organization of Output Codes in Multiclass Learning Problems

Wolfgang Utschick, W. Weichselberger

https://doi.org/10.1162/08997660151134334 | pp. 1065-1102

The best-known decomposition schemes of multiclass learning problems are one per class coding (OPC) and error-correcting output coding (ECOC). Both methods perform a prior decomposition, that is, before training of the classifier takes place. The impact of output codes on the inferred decision rules can be experienced only after learning. Therefore, we present a novel algorithm for the code design of multiclass learning problems. This algorithm applies a maximum-likelihood objective function in conjunction with the expectation-maximization (EM) algorithm. Minimizing the augmented objective function yields the optimal decomposition of the multiclass learning problem in two-class problems. Experimental results show the potential gain of the optimized output codes over OPC or ECOC methods.

59. Predictive Approaches for Choosing Hyperparameters in Gaussian Processes

S. Sundararajan, S. Sathiya Keerthi

https://doi.org/10.1162/08997660151134343 | pp. 1103-1118

Gaussian processes are powerful regression models specified by parameterized mean and covariance functions. Standard approaches to choose these parameters (known by the name hyperparameters) are maximum likelihood and maximum a posteriori. In this article, we propose and investigate predictive approaches based on Geisser's predictive sample reuse (PSR) methodology and the related Stone's cross-validation (CV) methodology. More specifically, we derive results for Geisser's surrogate predictive probability (GPP), Geisser's predictive mean square error (GPE), and the standard CV error and make a

comparative study. Within an approximation we arrive at the generalized cross-validation (GCV) and establish its relationship with the GPP and GPE approaches. These approaches are tested on a number of problems. Experimental results show that these approaches are strongly competitive with the existing approaches.

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60. Architecture-Independent Approximation of Functions

Vicente Ruiz de Angulo, Carme Torras

https://doi.org/10.1162/08997660151134352 | pp. 1119-1135

We show that minimizing the expected error of a feedforward network over a distribution of weights results in an approximation that tends to be independent of network size as the number of hidden units grows. This minimization can be easily performed, and the complexity of the resulting function implemented by the network is regulated by the variance of the weight distribution. For a fixed variance, there is a number of hidden units above which either the implemented function does not change or the change is slight and tends to zero as the size of the network grows. In sum, the control of the complexity depends on only the variance, not the architecture, provided it is large enough.

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61. Analyzing Holistic Parsers: Implications for Robust Parsing and Systematicity

Edward Kei Shiu Ho, Lai Wan Chan

https://doi.org/10.1162/08997660151134361 | pp. 1137-1170

Holistic parsers offer a viable alternative to traditional algorithmic parsers. They have good generalization performance and are robust inherently. In a holistic parser, parsing is achieved by mapping the connectionist representation of the input sentence to the connectionist representation of the target parse tree directly. Little prior knowledge of the underlying parsing mechanism thus needs to be assumed. However, it also makes holistic parsing difficult to understand. In this article, an analysis is presented for studying the operations of the confluent preorder parser (CPP). In the analysis, the CPP is viewed as a dynamical system, and holistic parsing is perceived as a sequence of state transitions through its state-space. The seemingly one-shot parsing mechanism can thus be elucidated as a step-by-step inference process, with the intermediate parsing decisions being reflected by the states visited during parsing. The study serves two purposes. First, it improves our understanding of how grammatical errors are corrected by the CPP. The occurrence of an error in a sentence will cause the CPP to deviate from the normal track that is followed when the original sentence is parsed. But as the remaining terminals are read, the two trajectories will gradually converge until finally the correct parse tree is produced. Second, it reveals that having systematic parse tree representations alone cannot guarantee good generalization performance in holistic parsing. More important, they need to be distributed in certain useful locations of the representational space. Sentences with similar trailing terminals should have their corresponding parse tree representations mapped to nearby locations in the representational space. The study provides concrete evidence that encoding the linearized parse trees as obtained via preorder traversal can satisfy such a requirement.

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62. The Computational Exploration of Visual Word Recognition in a Split Model

Richard Shillcock, Padraic Monaghan

https://doi.org/10.1162/08997660151134370 | pp. 1171-1198

We argue that the projection of the visual field to the cortex constrains and informs the modeling of visual word recognition. On the basis of anatomical and psychological evidence, we claim that the higher-level cognition involved in word recognition does not completely transcend initial foveal splitting. We present a schematic connectionist model of word recognition that instantiates the precise splitting of the visual field and the contralateral projection of the two hemifields. We explore the special nature of the exterior (i.e., first and last) letters of words in reading. The model produces the correct behavior spontaneously and robustly. We analyze this behavior of the model with respect to words and random patterns and conclude that the systematic division of the visual input has predictable, general informational consequences and is chiefly responsible for the exterior letters effect.

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63. Generalization in Interactive Networks: The Benefits of Inhibitory Competition and Hebbian Learning

Randall C. O'Reilly

https://doi.org/10.1162/08997660152002834 | pp. 1199-1241

Computational models in cognitive neuroscience should ideally use biological properties and powerful computational principles to produce behavior consistent with psychological findings. Error-driven backpropagation is computationally powerful and has proven useful for modeling a range of psychological data but is not biologically plausible. Several approaches to implementing backpropagation in a biologically plausible fashion converge on the idea of using bidirectional activation propagation in interactive networks to convey error signals. This article demonstrates two main points about these error-driven interactive networks: (1) they generalize poorly due to attractor dynamics that interfere with the network's ability to produce novel combinatorial representations systematically in response to novel inputs, and (2) this generalization problem can be remedied by adding two widely used mechanistic principles, inhibitory competition and Hebbian learning, that can be independently motivated for a variety of biological, psychological, and computational reasons. Simulations using the Leabra algorithm, which combines the generalized recirculation (GeneRec), biologically plausible, error-driven learning algorithm with inhibitory competition and Hebbian learning, show that these mechanisms can result in good generalization in interactive networks. These results support the general conclusion that cognitive neuroscience models that incorporate the core mechanistic principles of interactivity, inhibitory competition, and error-driven and Hebbian learning satisfy a wider range of biological, psychological, and computational constraints than models employing a subset of these principles.

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64. Optimal Smoothing in Visual Motion Perception

Rajesh P. N. Rao, David M. Eagleman, Terrence J. Sejnowski

https://doi.org/10.1162/08997660152002843 | pp. 1243-1253

When a flash is aligned with a moving object, subjects perceive the flash to lag behind the moving object. Two different models have been proposed to explain this "flash-lag" effect. In the motion extrapolation model, the visual system extrapolates the location of the moving object to counteract neural propagation delays, whereas in the latency difference model, it is hypothesized that moving objects are processed and perceived more quickly than flashed objects. However, recent psychophysical experiments suggest that neither of these interpretations is feasible (Eagleman & Sejnowski, 2000a, 2000b, 2000c), hypothesizing instead that the visual system uses data from the future of an event before committing to an interpretation. We formalize this idea in terms of the statistical framework of optimal smoothing and show that a model based on smoothing accounts for the shape of psychometric curves from a flash-lag experiment involving random reversals of motion direction. The smoothing model demonstrates how the visual system may enhance perceptual accuracy by relying not only on data from the past but also on data collected from the immediate future of an event.

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65. Rate Coding Versus Temporal Order Coding: What the Retinal Ganglion Cells Tell the Visual Cortex

Rufin Van Rullen, Simon J. Thorpe

https://doi.org/10.1162/08997660152002852 | pp. 1255-1283

It is often supposed that the messages sent to the visual cortex by the retinal ganglion cells are encoded by the mean firing rates observed on spike trains generated with a Poisson process. Using an information transmission approach, we evaluate the performances of two such codes, one based on the spike count and the other on the mean interspike interval, and compare the results with a rank order code, where the first ganglion cells to emit a spike are given a maximal weight. Our results show that the rate codes are far from optimal for fast information transmission and that the temporal structure of the spike train can be efficiently used to maximize the information transfer rate under conditions where each cell needs to fire only one spike.

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66. The Effects of Spike Frequency Adaptation and Negative Feedback on the Synchronization of Neural Oscillators

Bard Ermentrout, Matthew Pascal, Boris Gutkin

https://doi.org/10.1162/08997660152002861 | pp. 1285-1310

There are several different biophysical mechanisms for spike frequency adaptation observed in recordings from cortical neurons. The two most commonly used in modeling studies are a calcium-dependent potassium current I(ahp) and a slow voltage-dependent potassium current, I(m). We show that both of these have strong effects on the synchronization properties of excitatorily coupled neurons. Furthermore, we show that the reasons for these effects are different. We show through an analysis of some standard models, that the M-current adaptation alters the mechanism for repetitive firing, while the afterhyperpolarization adaptation works via shunting the incoming synapses. This latter mechanism applies with a network that has recurrent inhibition. The shunting behavior is captured in a simple two-variable reduced model that arises near certain types of bifurcations. A one-dimensional map is derived from the simplified model.

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67. A Unified Approach to the Study of Temporal, Correlational, and Rate Coding

Stefano Panzeri, Simon R. Schultz

https://doi.org/10.1162/08997660152002870 | pp. 1311-1349

We demonstrate that the information contained in the spike occurrence times of a population of neurons can be broken up into a series of terms, each reflecting something about potential coding mechanisms. This is possible in the coding regime in which few spikes are emitted in the relevant time window. This approach allows us to study the additional information contributed by spike timing beyond that present in the spike counts and to examine the contributions to the whole information of different statistical properties of spike trains, such as firing rates and correlation functions. It thus forms the basis for a new quantitative procedure for analyzing simultaneous multiple neuron recordings and provides theoretical constraints on neural coding strategies. We find a transition between two coding regimes, depending on the size of the relevant observation timescale. For time windows shorter than the timescale of the stimulus-induced response fluctuations, there exists a spike count coding phase, in which the purely temporal information is of third order in time. For time windows much longer than the characteristic timescale, there can be additional timing information of first order, leading to a temporal coding phase in which timing information may affect the instantaneous information rate. In this new framework, we study the relative contributions of the dynamic firing rate and correlation variables to the full temporal information, the interaction of signal and noise correlations in temporal coding, synergy between spikes and between cells, and the effect of refractoriness. We illustrate the utility of the technique by analyzing a few cells from the rat barrel cortex.

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68. Determination of Response Latency and Its Application to Normalization of Cross-Correlation Measures

Stuart N. Baker, George L. Gerstein

https://doi.org/10.1162/08997660152002889 | pp. 1351-1377

It is often of interest experimentally to assess how synchronization between two neurons changes following a stimulus or other behaviorally relevant marker. The joint peristimulus time histogram (JPSTH) achieves this, but assumes that changes in the cells' firing rate following the stimulus are stereotyped from one sweep to the next. Erroneous results can be generated if this is not the case. We here present a method to assess whether there are variations in response latency or amplitude from sweep to sweep. We then describe how the effects of response latency variation can be mitigated by realigning sweeps to their individual latencies. Three methods of detecting response latency are presented and their performance compared on simulated data. Finally, the effect on the JPSTH of sweep realignment using detected latencies is illustrated.

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69. Attractive Periodic Sets in Discrete-Time Recurrent Networks (with Emphasis on Fixed-Point Stability and Bifurcations in Two-Neuron Networks)

Peter Ti■o, Bill G. Horne, C. Lee Giles

https://doi.org/10.1162/08997660152002898 | pp. 1379-1414

We perform a detailed fixed-point analysis of two-unit recurrent neural networks with sigmoid-shaped transfer functions. Using geometrical arguments in the space of transfer function derivatives, we partition the

network state-space into distinct regions corresponding to stability types of the fixed points. Unlike in the previous studies, we do not assume any special form of connectivity pattern between the neurons, and all free parameters are allowed to vary. We also prove that when both neurons have excitatory self-connections and the mutual interaction pattern is the same (i.e., the neurons mutually inhibit or excite themselves), new attractive fixed points are created through the saddle-node bifurcation. Finally, for an N-neuron recurrent network, we give lower bounds on the rate of convergence of attractive periodic points toward the saturation values of neuron activations, as the absolute values of connection weights grow.

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70. Attractor Networks for Shape Recognition

Yali Amit, Massimo Mascaro

https://doi.org/10.1162/08997660152002906 | pp. 1415-1442

We describe a system of thousands of binary perceptrons with coarse-oriented edges as input that is able to recognize shapes, even in a context with hundreds of classes. The perceptrons have randomized feedforward connections from the input layer and form a recurrent network among themselves. Each class is represented by a prelearned attractor (serving as an associative hook) in the recurrent net corresponding to a randomly selected subpopulation of the perceptrons. In training, first the attractor of the correct class is activated among the perceptrons; then the visual stimulus is presented at the input layer. The feedforward connections are modified using field-dependent Hebbian learning with positive synapses, which we show to be stable with respect to large variations in feature statistics and coding levels and allows the use of the same threshold on all perceptrons. Recognition is based on only the visual stimuli. These activate the recurrent network, which is then driven by the dynamics to a sustained attractor state, concentrated in the correct class subset and providing a form of working memory. We believe this architecture is more transparent than standard feedforward two-layer networks and has stronger biological analogies.

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71. Estimating the Support of a High-Dimensional Distribution

Bernhard Schölkopf, John Platt, John Shawe■Taylor, Alex Smola, Robert C. Williamson

https://doi.org/10.1162/089976601750264965 | pp. 1443-1471

Suppose you are given some data set drawn from an underlying probability distribution P and you want to estimate a "simple" subset S of input space such that the probability that a test point drawn from P lies outside of S equals some a priori specified value between 0 and 1. We propose a method to approach this problem by trying to estimate a function f that is positive on S and negative on the complement. The functional form of f is given by a kernel expansion in terms of a potentially small subset of the training data; it is regularized by controlling the length of the weight vector in an associated feature space. The expansion coefficients are found by solving a quadratic programming problem, which we do by carrying out sequential optimization over pairs of input patterns. We also provide a theoretical analysis of the statistical performance of our algorithm. The algorithm is a natural extension of the support vector algorithm to the case of unlabeled data.

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72. Stationary Bumps in Networks of Spiking Neurons

Carlo R. Laing, Carson C. Chow

https://doi.org/10.1162/089976601750264974 | pp. 1473-1494

We examine the existence and stability of spatially localized "bumps" of neuronal activity in a network of spiking neurons. Bumps have been proposed in mechanisms of visual orientation tuning, the rat head direction system, and working memory. We show that a bump solution can exist in a spiking network provided the neurons fire asynchronously within the bump. We consider a parameter regime where the bump solution is bistable with an all-off state and can be initiated with a transient excitatory stimulus. We show that the activity profile matches that of a corresponding population rate model. The bump in a spiking network can lose stability through partial synchronization to either a traveling wave or the all-off state. This can occur if the synaptic timescale is too fast through a dynamical effect or if a transient excitatory pulse is applied to the network. A bump can thus be activated and deactivated with excitatory inputs that may have physiological relevance.

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73. Vergence Dynamics Predict Fixation Disparity

Saumil S. Patel, Baichuan Jiang, Haluk Ö■men

https://doi.org/10.1162/089976601750264983 | pp. 1495-1525

The neural origin of the steady-state vergence eye movement error, called binocular fixation disparity, is not well understood. Further, there has been no study that quantitatively relates the dynamics of the vergence system to its steady-state behavior, a critical test for the understanding of any oculomotor system. We investigate whether fixation disparity can be related to the dynamics of opponent convergence and divergence neural pathways. Using binocular eye movement recordings, we first show that opponent vergence pathways exhibit asymmetric angle-dependent gains. We then present a neural model that combines physiological properties of disparity-tuned cells and vergence premotor cells with the asymmetric gain properties of the opponent pathways. Quantitative comparison of the model predictions with our experimental data suggests that fixation disparity can arise when asymmetric opponent vergence pathways are driven by a distributed disparity code.

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74. Topographic Independent Component Analysis

Aapo Hyvärinen, Patrik O. Hoyer, Mika Inki

https://doi.org/10.1162/089976601750264992 | pp. 1527-1558

In ordinary independent component analysis, the components are assumed to be completely independent, and they do not necessarily have any meaningful order relationships. In practice, however, the estimated "independent" components are often not at all independent. We propose that this residual dependence structure could be used to define a topographic order for the components. In particular, a distance between two components could be defined using their higher-order correlations, and this distance could be used to create a topographic representation. Thus, we obtain a linear decomposition into approximately independent components, where the dependence of two components is approximated by the proximity of the components in the topographic representation.

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75. Blind Source Separation Using Temporal Predictability

James V. Stone

https://doi.org/10.1162/089976601750265009 | pp. 1559-1574

A measure of temporal predictability is defined and used to separate linear mixtures of signals. Given any set of statistically independent source signals, it is conjectured here that a linear mixture of those signals has the following property: the temporal predictability of any signal mixture is less than (or equal to) that of any of its component source signals. It is shown that this property can be used to recover source signals from a set of linear mixtures of those signals by finding an un-mixing matrix that maximizes a measure of temporal predictability for each recovered signal. This matrix is obtained as the solution to a generalized eigenvalue problem; such problems have scaling characteristics of O(N3), where N is the number of signal mixtures. In contrast to independent component analysis, the temporal predictability method requires minimal assumptions regarding the probability density functions of source signals. It is demonstrated that the method can separate signal mixtures in which each mixture is a linear combination of source signals with supergaussian, subgaussian, and gaussian probability density functions and on mixtures of voices and music.

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76. A Biologically Motivated Solution to the Cocktail Party Problem

Brian Sagi, Syrus C. Nemat-Nasser, Rex Kerr, Raja Hayek, Christopher Downing, Robert Hecht-Nielsen

https://doi.org/10.1162/089976601750265018 | pp. 1575-1602

We present a new approach to the cocktail party problem that uses a cortronic artificial neural network architecture (Hecht-Nielsen, 1998) as the front end of a speech processing system. Our approach is novel in three important respects. First, our method assumes and exploits detailed knowledge of the signals we wish to attend to in the cocktail party environment. Second, our goal is to provide preprocessing in advance of a pattern recognition system rather than to separate one or more of the mixed sources explicitly. Third, the neural network model we employ is more biologically feasible than are most other approaches to the cocktail party problem. Although the focus here is on the cocktail party problem, the method presented in this study can be applied to other areas of information processing.

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77. A Comparative Study of Feature-Salience Ranking Techniques

Wenli Wang, P. Jones, Derek Partridge

https://doi.org/10.1162/089976601750265027 | pp. 1603-1623

We assess the relative merits of a number of techniques designed to determine the relative salience of the elements of a feature set with respect to their ability to predict a category outcome-for example, which features of a character contribute most to accurate character recognition. A number of different neural-net-based techniques have been proposed (by us and others) in addition to a standard statistical technique, and we add a technique based on inductively generated decision trees. The salience of the features that compose a proposed set is an important problem to solve efficiently and effectively, not only for neural computing technology but also in order to provide a sound basis for any attempt to design an optimal computational system. The focus of this study is the efficiency and the effectiveness with which high-salience subsets of features can be identified in the context of ill-understood and potentially noisy

real-world data. Our two simple approaches, weight clamping using a neural network and feature ranking using a decision tree, generally provide a good, consistent ordering of features. In addition, linear correlation often works well.

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78. A Theory for Learning by Weight Flow on Stiefel-Grassman Manifold

Simone Fiori

https://doi.org/10.1162/089976601750265036 | pp. 1625-1647

Recently we introduced the concept of neural network learning on Stiefel-Grassman manifold for multilayer perceptron—like networks. Contributions of other authors have also appeared in the scientific literature about this topic. This article presents a general theory for it and illustrates how existing theories may be explained within the general framework proposed here.

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79. Online Model Selection Based on the Variational Bayes

Masa-aki Sato

https://doi.org/10.1162/089976601750265045 | pp. 1649-1681

The Bayesian framework provides a principled way of model selection. This framework estimates a probability distribution over an ensemble of models, and the prediction is done by averaging over the ensemble of models. Accordingly, the uncertainty of the models is taken into account, and complex models with more degrees of freedom are penalized. However, integration over model parameters is often intractable, and some approximation scheme is needed. Recently, a powerful approximation scheme, called the variational bayes (VB) method, has been proposed. This approach defines the free energy for a trial probability distribution, which approximates a joint posterior probability distribution over model parameters and hidden variables. The exact maximization of the free energy gives the true posterior distribution. The VB method uses factorized trial distributions. The integration over model parameters can be done analytically, and an iterative expectation-maximization-like algorithm, whose convergence is guaranteed, is derived. In this article, we derive an online version of the VB algorithm and prove its convergence by showing that it is a stochastic approximation for finding the maximum of the free energy. By combining sequential model selection procedures, the online VB method provides a fully online learning method with a model selection mechanism. In preliminary experiments using synthetic data, the online VB method was able to adapt the model structure to dynamic environments.

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80. Orientation, Scale, and Discontinuity as Emergent Properties of Illusory Contour Shape

Lance R. Williams, Karvel K. Thornber

https://doi.org/10.1162/08997660152469305 | pp. 1683-1711

A recent neural model of illusory contour formation is based on a distribution of natural shapes traced by particles moving with constant speed in directions given by Brownian motions. The input to that model consists of pairs of position and direction constraints, and the output consists of the distribution of contours joining all such pairs. In general, these contours will not be closed, and their distribution will not be

scale-invariant. In this article, we show how to compute a scale-invariant distribution of closed contours given position constraints alone and use this result to explain a well-known illusory contour effect.

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81. A Spike-Train Probability Model

Robert E. Kass, Valérie Ventura

https://doi.org/10.1162/08997660152469314 | pp. 1713-1720

Poisson processes usually provide adequate descriptions of the irregularity in neuron spike times after pooling the data across large numbers of trials, as is done in constructing the peristimulus time histogram. When probabilities are needed to describe the behavior of neurons within individual trials, however, Poisson process models are often inadequate. In principle, an explicit formula gives the probability density of a single spike train in great generality, but without additional assumptions, the firing-rate intensity function appearing in that formula cannot be estimated. We propose a simple solution to this problem, which is to assume that the time at which a neuron fires is determined probabilistically by, and only by, two quantities: the experimental clock time and the elapsed time since the previous spike. We show that this model can be fitted with standard methods and software and that it may used successfully to fit neuronal data.

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82. Orientation Tuning Properties of Simple Cells in Area V1 Derived from an Approximate Analysis of Nonlinear Neural Field Models

Thomas Wennekers

https://doi.org/10.1162/08997660152469323 | pp. 1721-1747

We present a general approximation method for the mathematical analysis of spatially localized steady-state solutions in nonlinear neural field models. These models comprise several layers of excitatory and inhibitory cells. Coupling kernels between and inside layers are assumed to be gaussian shaped. In response to spatially localized (i.e., tuned) inputs, such networks typically reveal stationary localized activity profiles in the different layers. Qualitative properties of these solutions, like response amplitudes and tuning widths, are approximated for a whole class of nonlinear rate functions that obey a power law above some threshold and that are zero below. A special case of these functions is the semilinear function, which is commonly used in neural field models. The method is then applied to models for orientation tuning in cortical simple cells: first, to the one-layer model with "difference of gaussians" connectivity kernel developed by Carandini and Ringach (1997) as an abstraction of the biologically detailed simulations of Somers, Nelson, and Sur (1995); second, to a two-field model comprising excitatory and inhibitory cells in two separate layers. Under certain conditions, both models have the same steady states. Comparing simulations of the field models and results derived from the approximation method, we find that the approximation well predicts the tuning behavior of the full model. Moreover, explicit formulas for approximate amplitudes and tuning widths in response to changing input strength are given and checked numerically. Comparing the network behavior for different nonlinearities, we find that the only rate function (from the class of functions under study) that leads to constant tuning widths and a linear increase of firing rates in response to increasing input is the semilinear function. For other nonlinearities, the qualitative network response depends on whether the model neurons operate in a convex (e.g., x(2)) or concave (e.g., sqrt(x)) regime of their rate function. In the first case, tuning gradually changes from input driven at low input strength (broad tuning strongly depending on the input and roughly linear amplitudes in response to input strength) to recurrently driven at moderate input strength

(sharp tuning, supralinear increase of amplitudes in response to input strength). For concave rate functions, the network reveals stable hysteresis between a state at low firing rates and a tuned state at high rates. This means that the network can "memorize" tuning properties of a previously shown stimulus. Sigmoid rate functions can combine both effects. In contrast to the Carandini-Ringach model, the two-field model further reveals oscillations with typical frequencies in the beta and gamma range, when the excitatory and inhibitory connections are relatively strong. This suggests a rhythmic modulation of tuning properties during cortical oscillations.

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83. Computational Design and Nonlinear Dynamics of a Recurrent Network Model of the Primary Visual Cortex*

Li Zhaoping

https://doi.org/10.1162/08997660152469332 | pp. 1749-1780

Recurrent interactions in the primary visual cortex make its output a complex nonlinear transform of its input. This transform serves preattentive visual segmentation, that is, autonomously processing visual inputs to give outputs that selectively emphasize certain features for segmentation. An analytical understanding of the nonlinear dynamics of the recurrent neural circuit is essential to harness its computational power. We derive requirements on the neural architecture, components, and connection weights of a biologically plausible model of the cortex such that region segmentation, figure-ground segregation, and contour enhancement can be achieved simultaneously. In addition, we analyze the conditions governing neural oscillations, illusory contours, and the absence of visual hallucinations. Many of our analytical techniques can be applied to other recurrent networks with translation-invariant neural and connection structures.

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84. A Hierarchical Dynamical Map as a Basic Frame for Cortical Mapping and Its Application to Priming

Osamu Hoshino, Satoru Inoue, Yoshiki Kashimori, Takeshi Kambara

https://doi.org/10.1162/08997660152469341 | pp. 1781-1810

A hierarchical dynamical map is proposed as the basic framework for sensory cortical mapping. To show how the hierarchical dynamical map works in cognitive processes, we applied it to a typical cognitive task known as priming, in which cognitive performance is facilitated as a consequence of prior experience. Prior to the priming task, the network memorizes a sensory scene containing multiple objects presented simultaneously using a hierarchical dynamical map. Each object is composed of different sensory features. The hierarchical dynamical map presented here is formed by random itinerancy among limit-cycle attractors into which these objects are encoded. Each limit-cycle attractor contains multiple point attractors into which elemental features belonging to the same object are encoded. When a feature stimulus is presented as a priming cue, the network state is changed from the itinerant state to a limit-cycle attractor relevant to the priming cue. After a short priming period, the network state reverts to the itinerant state. Under application of the test cue, consisting of some feature belonging to the object relevant to the priming cue and fragments of features belonging to others, the network state is changed to a limit-cycle attractor and finally to a point attractor relevant to the target feature. This process is considered as the identification of the target. The model consistently reproduces various observed results for priming processes such as the difference in identification time between cross-modality and within-modality priming tasks, the effect of interval between

priming cue and test cue on identification time, the effect of priming duration on the time, and the effect of repetition of the same priming task on neural activity.

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85. Dynamical Stability Conditions for Recurrent Neural Networks with Unsaturating Piecewise Linear Transfer Functions

Heiko Wersing, Wolf■Jürgen Beyn, Helge Ritter

https://doi.org/10.1162/08997660152469350 | pp. 1811-1825

We establish two conditions that ensure the nondivergence of additive recurrent networks with unsaturating piecewise linear transfer functions, also called linear threshold or semilinear transfer functions. As Hahnloser, Sarpeshkar, Mahowald, Douglas, and Seung (2000) showed, networks of this type can be efficiently built in silicon and exhibit the coexistence of digital selection and analog amplification in a single circuit. To obtain this behavior, the network must be multistable and nondivergent, and our conditions allow determining the regimes where this can be achieved with maximal recurrent amplification. The first condition can be applied to nonsymmetric networks and has a simple interpretation of requiring that the strength of local inhibition match the sum over excitatory weights converging onto a neuron. The second condition is restricted to symmetric networks, but can also take into account the stabilizing effect of nonlocal inhibitory interactions. We demonstrate the application of the conditions on a simple example and the orientation-selectivity model of Ben-Yishai, Lev Bar-Or, and Sompolinsky (1995). We show that the conditions can be used to identify in their model regions of maximal orientation-selective amplification and symmetry breaking.

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86. An Information-Based Neural Approach to Constraint Satisfaction

Henrik Jönsson, Bo Söderberg

https://doi.org/10.1162/08997660152469369 | pp. 1827-1838

A novel artificial neural network approach to constraint satisfaction problems is presented. Based on information-theoretical considerations, it differs from a conventional mean-field approach in the form of the resulting free energy. The method, implemented as an annealing algorithm, is numerically explored on a testbed of K-SAT problems. The performance shows a dramatic improvement over that of a conventional mean-field approach and is comparable to that of a state-of-the-art dedicated heuristic (GSAT+walk). The real strength of the method, however, lies in its generality. With minor modifications, it is applicable to arbitrary types of discrete constraint satisfaction problems.

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87. Recurrence Methods in the Analysis of Learning Processes

Shahar Mendelson, Israel Nelken

https://doi.org/10.1162/08997660152469378 | pp. 1839-1861

The goal of most learning processes is to bring a machine into a set of "correct" states. In practice, however, it may be difficult to show that the process enters this target set. We present a condition that ensures that the process visits the target set infinitely often almost surely. This condition is easy to verify and is true for many well-known learning rules. To demonstrate the utility of this method, we apply it to four types of

learning processes: the perceptron, learning rules governed by continuous energy functions, the Kohonen rule, and the committee machine.

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88. Subspace Information Criterion for Model Selection

Masashi Sugiyama, Hidemitsu Ogawa

https://doi.org/10.1162/08997660152469387 | pp. 1863-1889

The problem of model selection is considerably important for acquiring higher levels of generalization capability in supervised learning. In this article, we propose a new criterion for model selection, the subspace information criterion (SIC), which is a generalization of Mallows's C(L). It is assumed that the learning target function belongs to a specified functional Hilbert space and the generalization error is defined as the Hilbert space squared norm of the difference between the learning result function and target function. SIC gives an unbiased estimate of the generalization error so defined. SIC assumes the availability of an unbiased estimate of the target function and the noise covariance matrix, which are generally unknown. A practical calculation method of SIC for least-mean-squares learning is provided under the assumption that the dimension of the Hilbert space is less than the number of training examples. Finally, computer simulations in two examples show that SIC works well even when the number of training examples is small.

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89. Convergent Decomposition Techniques for Training RBF Neural Networks

C. Buzzi, Luigi Grippo, Marco Sciandrone

https://doi.org/10.1162/08997660152469396 | pp. 1891-1920

In this article we define globally convergent decomposition algorithms for supervised training of generalized radial basis function neural networks. First, we consider training algorithms based on the two-block decomposition of the network parameters into the vector of weights and the vector of centers. Then we define a decomposition algorithm in which the selection of the center locations is split into sequential minimizations with respect to each center, and we give a suitable criterion for choosing the centers that must be updated at each step. We prove the global convergence of the proposed algorithms and report the computational results obtained for a set of test problems.

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90. Errata

(authors unknown)

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No abstract available.

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91. Modeling Neuronal Assemblies: Theory and Implementation

Julian Eggert, J. Leo van Hemmen

https://doi.org/10.1162/089976601750399254 | pp. 1923-1974

Models that describe qualitatively and quantitatively the activity of entire groups of spiking neurons are becoming increasingly important for biologically realistic large-scale network simulations. At the systems and areas modeling level, it is necessary to switch the basic descriptional level from single spiking neurons to neuronal assemblies. In this article, we present and review work that allows a macroscopic description of the assembly activity. We show that such macroscopic models can be used to reproduce in a quantitatively exact manner the joint activity of groups of spike-response or integrate-and-fire neurons. We also show that integral as well as differential equation models of neuronal assemblies can be understood within a single framework, which allows a comparison with the commonly used assembly-averaged graded-response type of models. The presented framework thus enables the large-scale neural network modeler to implement networks using computational units beyond the single spiking neuron without losing much biological accuracy. This article explains the theoretical background as well as the capabilities and the implementation details of the assembly approach.

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92. On a Class of Support Vector Kernels Based on Frames in Function Hilbert Spaces

Junbin Gao, C.J. Harris, S.R. Gunn

https://doi.org/10.1162/089976601750399263 | pp. 1975-1994

There has been an increasing interest in kernel-based techniques, such as support vector techniques, regularization networks, and gaussian processes. There are inner relationships among those techniques, with the kernel function playing a central role. This article discusses a new class of kernel functions derived from the so-called frames in a function Hilbert space.

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93. Extraction of Specific Signals with Temporal Structure

Allan Kardec Barros, Andrzej Cichocki

https://doi.org/10.1162/089976601750399272 | pp. 1995-2003

In this work we develop a very simple batch learning algorithm for semiblind extraction of a desired source signal with temporal structure from linear mixtures. Although we use the concept of sequential blind extraction of sources and independent component analysis, we do not carry out the extraction in a completely blind manner; neither do we assume that sources are statistically independent. In fact, we show that the a priori information about the autocorrelation function of primary sources can be used to extract the desired signals (sources of interest) from their linear mixtures. Extensive computer simulations and real data application experiments confirm the validity and high performance of the proposed algorithm.

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94. Correlation Between Uncoupled Conductance-Based Integrate-and-Fire Neurons Due to Common and Synchronous Presynaptic Firing

Sybert Stroeve, Stan Gielen

https://doi.org/10.1162/089976601750399281 | pp. 2005-2029

We investigate the firing characteristics of conductance-based integrate-and-fire neurons and the correlation of firing for uncoupled pairs of neurons as a result of common input and synchronous firing of multiple

synaptic inputs. Analytical approximations are derived for the moments of the steady state potential and the effective time constant. We show that postsynaptic firing barely depends on the correlation between inhibitory inputs; only the inhibitory firing rate matters. In contrast, both the degree of synchrony and the firing rate of excitatory inputs are relevant. A coefficient of variation CV > 1 can be attained with low inhibitory firing rates and (Poisson-modulated) synchronized excitatory synaptic input, where both the number of presynaptic neurons in synchronous firing assemblies and the synchronous firing rate should be sufficiently large. The correlation in firing of a pair of uncoupled neurons due to common excitatory input is initially increased for increasing firing rates of independent inhibitory inputs but decreases for large inhibitory firing rates. Common inhibitory input to a pair of uncoupled neurons barely induces correlated firing, but amplifies the effect of common excitation. Synchronous firing assemblies in the common input further enhance the correlation and are essential to attain experimentally observed correlation values. Since uncorrelated common input (i.e., common input by neurons, which do not fire in synchrony) cannot induce sufficient postsynaptic correlation, we conclude that lateral couplings are essential to establish clusters of synchronously firing neurons.

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95. Attention Modulation of Neural Tuning Through Peak and Base Rate

Hiroyuki Nakahara, Si Wu, ■■■-■■■ ■■■■■

https://doi.org/10.1162/089976601750399290 | pp. 2031-2047

This study investigates the influence of attention modulation on neural tuning functions. It has been shown in experiments that attention modulation alters neural tuning curves. Attention has been considered at least to serve to resolve limiting capacities and to increase the sensitivity to attended stimulus, while the exact functions of attention are still under debate. Inspired by recent experimental results on attention modulation, we investigate the influence of changes in the height and base rate of the tuning curve on the encoding accuracy, using the Fisher information. Under an assumption of stimulus-conditional independence of neural responses, we derive explicit conditions that determine when the height and base rate should be increased or decreased to improve encoding accuracy. Notably, a decrease in the tuning height and base rate can improve the encoding accuracy in some cases. Our theoretical results can predict the effective size of attention modulation on the neural population with respect to encoding accuracy. We discuss how our method can be used quantitatively to evaluate different aspects of attention function.

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96. Democratic Integration: Self-Organized Integration of Adaptive Cues

Jochen Triesch, Christoph von der Malsburg

https://doi.org/10.1162/089976601750399308 | pp. 2049-2074

Sensory integration or sensor fusion -- the integration of information from different modalities, cues, or sensors -- is among the most fundamental problems of perception in biological and artificial systems. We propose a new architecture for adaptively integrating different cues in a self-organized manner. In Democratic Integration different cues agree on a result, and each cue adapts toward the result agreed on. In particular, discordant cues are quickly suppressed and recalibrated, while cues having been consistent with the result in the recent past are given a higher weight in the future. The architecture is tested in a face tracking scenario. Experiments show its robustness with respect to sudden changes in the environment as long as the changes disrupt only a minority of cues at the same time, although all cues may be disrupted at

one time or another.

97. An Autoassociative Neural Network Model of Paired-Associate Learning

Daniel S. Rizzuto, Michael J. Kahana

https://doi.org/10.1162/089976601750399317 | pp. 2075-2092

Hebbian heteroassociative learning is inherently asymmetric. Storing a forward association, from item A to item B, enables recall of B (given A), but does not permit recall of A (given B). Recurrent networks can solve this problem by associating A to B and B back to A. In these recurrent networks, the forward and backward associations can be differentially weighted to account for asymmetries in recall performance. In the special case of equal strength forward and backward weights, these recurrent networks can be modeled as a single autoassociative network where A and B are two parts of a single, stored pattern. We analyze a general, recurrent neural network model of associative memory and examine its ability to fit a rich set of experimental data on human associative learning. The model fits the data significantly better when the forward and backward storage strengths are highly correlated than when they are less correlated. This network-based analysis of associative learning supports the view that associations between symbolic elements are better conceptualized as a blending of two ideas into a single unit than as separately modifiable forward and backward associations linking representations in memory.

98. Simple Recurrent Networks Learn Context-Free and Context-Sensitive Languages by Counting

Paul Rodriguez

https://doi.org/10.1162/089976601750399326 | pp. 2093-2118

It has been shown that if a recurrent neural network (RNN) learns to process a regular language, one can extract a finite-state machine (FSM) by treating regions of phase-space as FSM states. However, it has also been shown that one can construct an RNN to implement Turing machines by using RNN dynamics as counters. But how does a network learn languages that require counting? Rodriguez, Wiles, and Elman (1999) showed that a simple recurrent network (SRN) can learn to process a simple context-free language (CFL) by counting up and down. This article extends that to show a range of language tasks in which an SRN develops solutions that not only count but also copy and store counting information. In one case, the network stores information like an explicit storage mechanism. In other cases, the network stores information more indirectly in trajectories that are sensitive to slight displacements that depend on context. In this sense, an SRN can learn analog computation as a set of interdependent counters. This demonstrates how SRNs may be an alternative psychological model of language or sequence processing.

99. Training v-Support Vector Classifiers: Theory and Algorithms

Chih-Chung Chang, Chih■Jen Lin

https://doi.org/10.1162/089976601750399335 | pp. 2119-2147

The nu-support vector machine (nu-SVM) for classification proposed by Schölkopf, Smola, Williamson, and Bartlett (2000) has the advantage of using a parameter nu on controlling the number of support vectors. In

this article, we investigate the relation between nu-SVM and C-SVM in detail. We show that in general they are two different problems with the same optimal solution set. Hence, we may expect that many numerical aspects of solving them are similar. However, compared to regular C-SVM, the formulation of nu-SVM is more complicated, so up to now there have been no effective methods for solving large-scale nu-SVM. We propose a decomposition method for nu-SVM that is competitive with existing methods for C-SVM. We also discuss the behavior of nu-SVM by some numerical experiments.

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100. A Tighter Bound for Graphical Models

M.A.R. Leisink, Hilbert J. Kappen

https://doi.org/10.1162/089976601750399344 | pp. 2149-2171

We present a method to bound the partition function of a Boltzmann machine neural network with any odd-order polynomial. This is a direct extension of the mean-field bound, which is first order. We show that the third-order bound is strictly better than mean field. Additionally, we derive a third-order bound for the likelihood of sigmoid belief networks. Numerical experiments indicate that an error reduction of a factor of two is easily reached in the region where expansion-based approximations are useful.

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101. Correctness of Belief Propagation in Gaussian Graphical Models of Arbitrary Topology

Yair Weiss, William T. Freeman

https://doi.org/10.1162/089976601750541769 | pp. 2173-2200

Graphical models, such as Bayesian networks and Markov random fields, represent statistical dependencies of variables by a graph. Local "belief propagation" rules of the sort proposed by Pearl (1988) are guaranteed to converge to the correct posterior probabilities in singly connected graphs. Recently, good performance has been obtained by using these same rules on graphs with loops, a method we refer to as loopy belief propagation. Perhaps the most dramatic instance is the near Shannon-limit performance of "Turbo codes," whose decoding algorithm is equivalent to loopy propagation. Except for the case of graphs with a single loop, there has been little theoretical understanding of loopy propagation. Here we analyze belief propagation in networks with arbitrary topologies when the nodes in the graph describe jointly gaussian random variables. We give an analytical formula relating the true posterior probabilities with those calculated using loopy propagation. We give sufficient conditions for convergence and show that when belief propagation converges, it gives the correct posterior means for all graph topologies, not just networks with a single loop. These results motivate using the powerful belief propagation algorithm in a broader class of networks and help clarify the empirical performance results.

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102. MOSAIC Model for Sensorimotor Learning and Control

Masahiko Haruno, Daniel M. Wolpert, Mitsuo Kawato

https://doi.org/10.1162/089976601750541778 | pp. 2201-2220

Humans demonstrate a remarkable ability to generate accurate and appropriate motor behavior under many different and often uncertain environmental conditions. We previously proposed a new modular architecture,

the modular selection and identification for control (MOSAIC) model, for motor learning and control based on multiple pairs of forward (predictor) and inverse (controller) models. The architecture simultaneously learns the multiple inverse models necessary for control as well as how to select the set of inverse models appropriate for a given environment. It combines both feedforward and feedback sensorimotor information so that the controllers can be selected both prior to movement and subsequently during movement. This article extends and evaluates the MOSAIC architecture in the following respects. The learning in the architecture was implemented by both the original gradient-descent method and the expectation-maximization (EM) algorithm. Unlike gradient descent, the newly derived EM algorithm is robust to the initial starting conditions and learning parameters. Second, simulations of an object manipulation task prove that the architecture can learn to manipulate multiple objects and switch between them appropriately. Moreover, after learning, the model shows generalization to novel objects whose dynamics lie within the polyhedra of already learned dynamics. Finally, when each of the dynamics is associated with a particular object shape, the model is able to select the appropriate controller before movement execution. When presented with a novel shape-dynamic pairing, inappropriate activation of modules is observed followed by on-line correction.

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103. Spike-Timing-Dependent Hebbian Plasticity as Temporal Difference Learning

Rajesh P. N. Rao, Terrence J. Sejnowski

https://doi.org/10.1162/089976601750541787 | pp. 2221-2237

A spike-timing-dependent Hebbian mechanism governs the plasticity of recurrent excitatory synapses in the neocortex: synapses that are activated a few milliseconds before a postsynaptic spike are potentiated, while those that are activated a few milliseconds after are depressed. We show that such a mechanism can implement a form of temporal difference learning for prediction of input sequences. Using a biophysical model of a cortical neuron, we show that a temporal difference rule used in conjunction with dendritic backpropagating action potentials reproduces the temporally asymmetric window of Hebbian plasticity observed physio-logically. Furthermore, the size and shape of the window vary with the distance of the synapse from the soma. Using a simple example, we show how a spike-timing-based temporal difference learning rule can allow a network of neocortical neurons to predict an input a few milliseconds before the input's expected arrival.

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104. Analysis and Neuronal Modeling of the Nonlinear Characteristics of a Local Cardiac Reflex in the Rat

Rajanikanth Vadigepalli, Francis J. Doyle, James S. Schwaber

https://doi.org/10.1162/089976601750541796 | pp. 2239-2271

Previous experimental results have suggested the existence of a local cardiac reflex in the rat. In this study, the putative role of such a local reflex in cardiovascular regulation is quantitatively analyzed. A model for the local reflex is developed from anatomical experimental results and physiological data in the literature. Using this model, a systems-level analysis is conducted. Simulation results indicate that the neuromodulatory mechanism of the local reflex attenuates the nonlinearity of the relationship between cardiac vagal drive and arterial pressure. This behavior is characterized through coherence analysis. Furthermore, the modulation of

phase-related characteristics of the cardiovascular system is suggested as a plausible mechanism for the nonlinear attenuation. Based on these results, it is plausible that the functional role of the local reflex is highly robust nonlinear compensation at the heart, which results in less complex dynamics in other parts of the reflex.

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105. Evaluating Auditory Performance Limits: I. One-Parameter Discrimination Using a Computational Model for the Auditory Nerve

Michael G. Heinz, H. Steven Colburn, Laurel H. Carney

https://doi.org/10.1162/089976601750541804 | pp. 2273-2316

A method for calculating psychophysical performance limits based on stochastic neural responses is introduced and compared to previous analytical methods for evaluating auditory discrimination of tone frequency and level. The method uses signal detection theory and a computational model for a population of auditory nerve (AN) fiber responses. The use of computational models allows predictions to be made over a wider parameter range and with more complete descriptions of AN responses than in analytical models. Performance based on AN discharge times (all-information) is compared to performance based only on discharge counts (rate-place). After the method is verified over the range of parameters for which previous analytical models are applicable, the parameter space is then extended. For example, a computational model of AN activity that extends to high frequencies is used to explore the common belief that rate-place information is responsible for frequency encoding at high frequencies due to the rolloff in AN phase locking above 2 kHz. This rolloff is thought to eliminate temporal information at high frequencies. Contrary to this belief, results of this analysis show that rate-place predictions for frequency discrimination are inconsistent with human performance in the dependence on frequency for high frequencies and that there is significant temporal information in the AN up to at least 10 kHz. In fact, the all-information predictions match the functional dependence of human performance on frequency, although optimal performance is much better than human performance. The use of computational AN models in this study provides new constraints on hypotheses of neural encoding of frequency in the auditory system; however, the method is limited to simple tasks with deterministic stimuli. A companion article in this issue ("Evaluating Auditory Performance Limits: II") describes an extension of this approach to more complex tasks that include random variation of one parameter, for example, random-level variation, which is often used in psychophysics to test neural encoding hypotheses.

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106. Evaluating Auditory Performance Limits: II. One-Parameter Discrimination with Random-Level Variation

Michael G. Heinz, H. Steven Colburn, Laurel H. Carney

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Previous studies have combined analytical models of stochastic neural responses with signal detection theory (SDT) to predict psychophysical performance limits; however, these studies have typically been limited to simple models and simple psychophysical tasks. A companion article in this issue ("Evaluating Auditory Performance Limits: I") describes an extension of the SDT approach to allow the use of computational models that provide more accurate descriptions of neural responses. This article describes an extension to more complex psychophysical tasks. A general method is presented for evaluating

psychophysical performance limits for discrimination tasks in which one stimulus parameter is randomly varied. Psychophysical experiments often randomly vary a single parameter in order to restrict the cues that are available to the subject. The method is demonstrated for the auditory task of random-level frequency discrimination using a computational auditory nerve (AN) model. Performance limits based on AN discharge times (all-information) are compared to performance limits based only on discharge counts (rate place). Both decision models are successful in predicting that random-level variation has no effect on performance in quiet, which is the typical result in psychophysical tasks with random-level variation. The distribution of information across the AN population provides insight into how different types of AN information can be used to avoid the influence of random-level variation. The rate-place model relies on comparisons between fibers above and below the tone frequency (i.e., the population response), while the all-information model does not require such across-fiber comparisons. Frequency discrimination with random-level variation in the presence of high-frequency noise is also simulated. No effect is predicted for all-information, consistent with the small effect in human performance; however, a large effect is predicted for rate-place in noise with random-level variation.

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107. Adaptive Algorithm for Blind Separation from Noisy Time-Varying Mixtures

V. Koivunen, M. Enescu, Erkki Oja

https://doi.org/10.1162/089976601750541822 | pp. 2339-2357

This article addresses the problem of blind source separation from time-varying noisy mixtures using a state variable model and recursive estimation. An estimate of each source signal is produced real time at the arrival of new observed mixture vector. The goal is to perform the separation and attenuate noise simultaneously, as well as to adapt to changes that occur in the mixing system. The observed data are projected along the eigenvectors in signal subspace. The subspace is tracked real time. Source signals are modeled using low-order AR (autoregressive) models, and noise is attenuated by trading off between the model and the information provided by measurements. The type of zero-memory nonlinearity needed in separation is determined on-line. Predictor-corrector filter structures are proposed, and their performance is investigated in simulation using biomedical and communications signals at different noise levels and a time-varying mixing system. In quantitative comparison to other widely used methods, significant improvement in output signal-to-noise ratio is achieved.

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108. Robust Full Bayesian Learning for Radial Basis Networks

Christophe Andrieu, Nando de Freitas, Arnaud Doucet

https://doi.org/10.1162/089976601750541831 | pp. 2359-2407

We propose a hierarchical full Bayesian model for radial basis networks. This model treats the model dimension (number of neurons), model parameters, regularization parameters, and noise parameters as unknown random variables. We develop a reversible-jump Markov chain Monte Carlo (MCMC) method to perform the Bayesian computation. We find that the results obtained using this method are not only better than the ones reported previously, but also appear to be robust with respect to the prior specification. In addition, we propose a novel and computationally efficient reversible-jump MCMC simulated annealing algorithm to optimize neural networks. This algorithm enables us to maximize the joint posterior distribution

of the network parameters and the number of basis function. It performs a global search in the joint space of the parameters and number of parameters, thereby surmounting the problem of local minima to a large extent. We show that by calibrating the full hierarchical Bayesian prior, we can obtain the classical Akaike information criterion, Bayesian information criterion, and minimum description length model selection criteria within a penalized likelihood framework. Finally, we present a geometric convergence theorem for the algorithm with homogeneous transition kernel and a convergence theorem for the reversible-jump MCMC simulated annealing method.

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109. Predictability, Complexity, and Learning

William Bialek, Ilya Nemenman, Naftali Tishby

https://doi.org/10.1162/089976601753195969 | pp. 2409-2463

We define predictive information I(pred)(T) as the mutual information between the past and the future of a time series. Three qualitatively different behaviors are found in the limit of large observation times T:I(pred)(T) can remain finite, grow logarithmically, or grow as a fractional power law. If the time series allows us to learn a model with a finite number of parameters, then I(pred)(T) grows logarithmically with a coefficient that counts the dimensionality of the model space. In contrast, power-law growth is associated, for example, with the learning of infinite parameter (or nonparametric) models such as continuous functions with smoothness constraints. There are connections between the predictive information and measures of complexity that have been defined both in learning theory and the analysis of physical systems through statistical mechanics and dynamical systems theory. Furthermore, in the same way that entropy provides the unique measure of available information consistent with some simple and plausible conditions, we argue that the divergent part of I(pred)(T) provides the unique measure for the complexity of dynamics underlying a time series. Finally, we discuss how these ideas may be useful in problems in physics, statistics, and biology.

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110. Dendritic Subunits Determined by Dendritic Morphology

K. A. Lindsay, J. M. Ogden, J.R. Rosenberg

https://doi.org/10.1162/089976601753195978 | pp. 2465-2476

A theoretical framework is presented in which arbitrarily branched dendritic structures with nonhomogeneous membrane properties and nonuniform geometry can be transformed into an equivalent unbranched structure (equivalent cable). Rall's equivalent cylinder is seen to be one part of the equivalent cable in the special case of dendrites satisfying the Rall criteria. The relation between the branched dendrite and its equivalent unbranched representation is uniquely defined by an invertible mapping that connects configurations of inputs on the branched structure with those on the unbranched structure, and conversely. This mapping provides a new definition of dendritic subunit and provides a mechanism for characterizing local and nonlocal signal processing within dendritic structures.

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111. Computing the Optimally Fitted Spike Train for a Synapse

Thomas Natschläger, Wolfgang Maass

https://doi.org/10.1162/089976601753195987 | pp. 2477-2494

Experimental data have shown that synapses are heterogeneous: different synapses respond with different sequences of amplitudes of postsynaptic responses to the same spike train. Neither the role of synaptic dynamics itself nor the role of the heterogeneity of synaptic dynamics for computations in neural circuits is well understood. We present in this article two computational methods that make it feasible to compute for a given synapse with known synaptic parameters the spike train that is optimally fitted to the synapse in a certain sense. With the help of these methods, one can compute, for example, the temporal pattern of a spike train (with a given number of spikes) that produces the largest sum of postsynaptic responses for a specific synapse. Several other applications are also discussed. To our surprise, we find that most of these optimally fitted spike trains match common firing patterns of specific types of neurons that are discussed in the literature. Hence, our analysis provides a possible functional explanation for the experimentally observed regularity in the combination of specific types of synapses with specific types of neurons in neural circuits.

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112. Period Focusing Induced by Network Feedback in Populations of Noisy Integrate-and-Fire Neurons

Francisco B. Rodr■■guez, Alberto Suárez, Vicente López

https://doi.org/10.1162/089976601753195996 | pp. 2495-2516

The population dynamics of an ensemble of nonleaky integrate-and-fire stochastic neurons is studied. The model selected allows for a detailed analysis of situations where noise plays a dominant role. Simulations in a regime with weak to moderate interactions show that a mechanism of excitatory message interchange among the neurons leads to a decrease in the firing period dispersion of the individual units. The dispersion reduction observed is larger than what would be expected from the decrease in the period. This "period focusing" is explained using a mean-field model. It is a dynamical effect that arises from the progressive decrease of the effective firing threshold as a result of the messages received by each unit from the rest of the population. A back-of-the-envelope formula to calculate this nontrivial dispersion reduction and a simple geometrical description of the effect are also provided.

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113. A Variational Method for Learning Sparse and Overcomplete Representations

Mark Girolami

https://doi.org/10.1162/089976601753196003 | pp. 2517-2532

An expectation-maximization algorithm for learning sparse and overcomplete data representations is presented. The proposed algorithm exploits a variational approximation to a range of heavy-tailed distributions whose limit is the Laplacian. A rigorous lower bound on the sparse prior distribution is derived, which enables the analytic marginalization of a lower bound on the data likelihood. This lower bound enables the development of an expectation-maximization algorithm for learning the overcomplete basis vectors and inferring the most probable basis coefficients.

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114. Random Embedding Machines for Pattern Recognition

Yoram Baram

https://doi.org/10.1162/089976601753196012 | pp. 2533-2548

Real classification problems involve structured data that can be essentially grouped into a relatively small number of clusters. It is shown that, under a local clustering condition, a set of points of a given class, embedded in binary space by a set of randomly parameterized surfaces, is linearly separable from other classes, with arbitrarily high probability. We call such a data set a local relative cluster. The size of the embedding set is shown to be inversely proportional to the squared local clustering degree. A simple parameterization by embedding hyperplanes, implementing a voting system, results in a random reduction of the nearest-neighbor method and leads to the separation of multicluster data by a network with two internal layers. This represents a considerable reduction of the learning problem with respect to known techniques, resolving a long-standing question on the complexity of random embedding. Numerical tests show that the proposed method performs as well as state-of the-art methods and in a small fraction of the time.

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115. Manifold Stochastic Dynamics for Bayesian Learning

Mark Zlochin, Yoram Baram

https://doi.org/10.1162/089976601753196021 | pp. 2549-2572

We propose a new Markov Chain Monte Carlo algorithm, which is a generalization of the stochastic dynamics method. The algorithm performs exploration of the state-space using its intrinsic geometric structure, which facilitates efficient sampling of complex distributions. Applied to Bayesian learning in neural networks, our algorithm was found to produce results comparable to the best state-of-the-art method while consuming considerably less time.

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116. Resampling Method for Unsupervised Estimation of Cluster Validity

Erel Levine, Eytan Domany

https://doi.org/10.1162/089976601753196030 | pp. 2573-2593

We introduce a method for validation of results obtained by clustering analysis of data. The method is based on resampling the available data. A figure of merit that measures the stability of clustering solutions against resampling is introduced. Clusters that are stable against resampling give rise to local maxima of this figure of merit. This is presented first for a one-dimensional data set, for which an analytic approximation for the figure of merit is derived and compared with numerical measurements. Next, the applicability of the method is demonstrated for higher-dimensional data, including gene microarray expression data.

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117. The Whitney Reduction Network: A Method for Computing Autoassociative Graphs

D.S. Broomhead, Michael Kirby

https://doi.org/10.1162/089976601753196049 | pp. 2595-2616

This article introduces a new architecture and associated algorithms ideal for implementing the dimensionality reduction of an m-dimensional manifold initially residing in an n-dimensional Euclidean space where n >> m. Motivated by Whitney's embedding theorem, the network is capable of training the identity mapping employing the idea of the graph of a function. In theory, a reduction to a dimension d that retains the differential structure of the original data may be achieved for some d < or = 2m + 1. To implement this network, we propose the idea of a good-projection, which enhances the generalization capabilities of the network, and an adaptive secant basis algorithm to achieve it. The effect of noise on this procedure is also considered. The approach is illustrated with several examples.

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118. Enhanced 3D Shape Recovery Using the Neural-Based Hybrid Reflectance Model

Siu-Yeung Cho, Tommy W. S. Chow

https://doi.org/10.1162/089976601753196058 | pp. 2617-2637

It is known that most real surfaces usually are neither perfectly Lambertian model nor ideally specular model; rather, they are formed by the hybrid structure of these two models. This hybrid reflectance model still suffers from the noise, strong specular, and unknown reflectivity conditions. In this article, these limitations are addressed, and a new neural-based hybrid reflectance model is proposed. The goal of this method is to optimize a proper reflectance model by learning the weight and parameters of the hybrid structure of feedforward neural networks and radial basis function networks and to recover the 3D object shape by the shape from shading technique with this resulting model. Experimental results, including synthetic and real images, were performed to demonstrate the performance of the proposed reflectance model in the case of different specular effects and noise environments.

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119. Synchronization of the Neural Response to Noisy Periodic Synaptic Input

Anthony N. Burkitt, Graeme M. Clark

https://doi.org/10.1162/089976601317098475 | pp. 2639-2672

The timing information contained in the response of a neuron to noisy periodic synaptic input is analyzed for the leaky integrate-and-fire neural model. We address the question of the relationship between the timing of the synaptic inputs and the output spikes. This requires an analysis of the interspike interval distribution of the output spikes, which is obtained in the gaussian approximation. The conditional output spike density in response to noisy periodic input is evaluated as a function of the initial phase of the inputs. This enables the phase transition matrix to be calculated, which relates the phase at which the output spike is generated to the initial phase of the inputs. The interspike interval histogram and the period histogram for the neural response to ongoing periodic input are then evaluated by using the leading eigenvector of this phase transition matrix. The synchronization index of the output spikes is found to increase sharply as the inputs become synchronized. This enhancement of synchronization is most pronounced for large numbers of inputs and lower frequencies of modulation and also for rates of input near the critical input rate. However, the mutual information between the input phase of the stimulus and the timing of output spikes is found to decrease at low input rates as the number of inputs increases. The results show close agreement with those obtained from numerical simulations for large numbers of inputs.

120. Specification of Training Sets and the Number of Hidden Neurons for Multilayer Perceptrons

L. S. Camargo, Takashi Yoneyama

https://doi.org/10.1162/089976601317098484 | pp. 2673-2680

This work concerns the selection of input-output pairs for improved training of multilayer perceptrons, in the context of approximation of univariate real functions. A criterion for the choice of the number of neurons in the hidden layer is also provided. The main idea is based on the fact that Chebyshev polynomials can provide approximations to bounded functions up to a prescribed tolerance, and, in turn, a polynomial of a certain order can be fitted with a three-layer perceptron with a prescribed number of hidden neurons. The results are applied to a sensor identification example.

121. Spotting Neural Spike Patterns Using an Adversary Background Model

Itay Gat, Naftali Tishby

https://doi.org/10.1162/089976601317098493 | pp. 2681-2708

The detection of a specific stochastic pattern embedded in an unknown background noise is a difficult pattern recognition problem, encountered in many applications such as word spotting in speech. A similar problem emerges when trying to detect a multineural spike pattern in a single electrical recording, embedded in the complex cortical activity of a behaving animal. Solving this problem is crucial for the identification of neuronal code words with specific meaning. The technical difficulty of this detection is due to the lack of a good statistical model for the background activity, which rapidly changes with the recording conditions and activity of the animal. This work introduces the use of an adversary background model. This model assumes that the background "knows" the pattern sought, up to a first-order statistics, and this "knowledge" creates a background composed of all the permutations of our pattern. We show that this background model is tightly connected to the type-based information-theoretic approach. Furthermore, we show that computing the likelihood ratio is actually decomposing the log-likelihood distribution according to types of the empirical counts. We demonstrate the application of this method for detection of the reward patterns in the basal ganglia of behaving monkeys, yielding some unexpected biological results.

122. Intrinsic Stabilization of Output Rates by Spike-Based Hebbian Learning

Richard Kempter, Wulfram Gerstner, J. Leo van Hemmen

https://doi.org/10.1162/089976601317098501 | pp. 2709-2741

We study analytically a model of long-term synaptic plasticity where synaptic changes are triggered by presynaptic spikes, postsynaptic spikes, and the time differences between presynaptic and postsynaptic spikes. The changes due to correlated input and output spikes are quantified by means of a learning window. We show that plasticity can lead to an intrinsic stabilization of the mean firing rate of the postsynaptic neuron. Subtractive normalization of the synaptic weights (summed over all presynaptic inputs converging on a postsynaptic neuron) follows if, in addition, the mean input rates and the mean input correlations are identical at all synapses. If the integral over the learning window is positive, firing-rate stabilization requires a non-Hebbian component, whereas such a component is not needed if the integral of

the learning window is negative. A negative integral corresponds to anti-Hebbian learning in a model with slowly varying firing rates. For spike-based learning, a strict distinction between Hebbian and anti-Hebbian rules is questionable since learning is driven by correlations on the timescale of the learning window. The correlations between presynaptic and postsynaptic firing are evaluated for a piecewise-linear Poisson model and for a noisy spiking neuron model with refractoriness. While a negative integral over the learning window leads to intrinsic rate stabilization, the positive part of the learning window picks up spatial and temporal correlations in the input.

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123. Information Transfer Between Rhythmically Coupled Networks: Reading the Hippocampal Phase Code

Ole Jensen

https://doi.org/10.1162/089976601317098510 | pp. 2743-2761

There are numerous reports on rhythmic coupling between separate brain networks. It has been proposed that this rhythmic coupling indicates exchange of information. So far, few computational models have been proposed that explore this principle and its potential computational benefits. Recent results on hippocampal place cells of the rat provide new insight; it has been shown that information about space is encoded by the firing of place cells with respect to the phase of the ongoing theta rhythm. This principle is termed phase coding and suggests that upcoming locations (predicted by the hippocampus) are encoded by cells firing late in the theta cycle, whereas current location is encoded by early firing in the theta cycle. A network reading the hippocampal output must inevitably also receive an oscillatory theta input in order to decipher the phase-coded firing patterns. In this article, I propose a simple physiologically plausible mechanism implemented as an oscillatory network that can decode the hippocampal output. By changing only the phase of the theta input to the decoder, qualitatively different information is transferred: the theta phase determines whether representations of current or upcoming locations are read by the decoder. The proposed mechanism provides a computational principle for information transfer between oscillatory networks and might generalize to brain networks beyond the hippocampal region.

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124. Gaussian Process Approach to Spiking Neurons for Inhomogeneous Poisson Inputs

Ken■ichi Amemori, Shin Ishii

https://doi.org/10.1162/089976601317098529 | pp. 2763-2797

This article presents a new theoretical framework to consider the dynamics of a stochastic spiking neuron model with general membrane response to input spike. We assume that the input spikes obey an inhomogeneous Poisson process. The stochastic process of the membrane potential then becomes a gaussian process. When a general type of the membrane response is assumed, the stochastic process becomes a Markov-gaussian process. We present a calculation method for the membrane potential density and the firing probability density. Our new formulation is the extension of the existing formulation based on diffusion approximation. Although the single Markov assumption of the diffusion approximation simplifies the stochastic process analysis, the calculation is inaccurate when the stochastic process involves a multiple Markov property. We find that the variation of the shape of the membrane response, which has often been ignored in existing stochastic process studies, significantly affects the firing probability. Our approach can

consider the reset effect, which has been difficult to deal with by analysis based on the first passage time density.

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125. Dual Information Representation with Stable Firing Rates and Chaotic Spatiotemporal Spike Patterns in a Neural Network Model

Osamu Araki, Kazuyuki Aihara

https://doi.org/10.1162/089976601317098538 | pp. 2799-2822

Although various means of information representation in the cortex have been considered, the fundamental mechanism for such representation is not well understood. The relation between neural network dynamics and properties of information representation needs to be examined. We examined spatial pattern properties of mean firing rates and spatiotemporal spikes in an interconnected spiking neural network model. We found that whereas the spatiotemporal spike patterns are chaotic and unstable, the spatial patterns of mean firing rates (SPMFR) are steady and stable, reflecting the internal structure of synaptic weights. Interestingly, the chaotic instability contributes to fast stabilization of the SPMFR. Findings suggest that there are two types of network dynamics behind neuronal spiking: internally-driven dynamics and externally driven dynamics. When the internally driven dynamics dominate, spikes are relatively more chaotic and independent of external inputs; the SPMFR are steady and stable. When the externally driven dynamics dominate, the spiking patterns are relatively more dependent on the spatiotemporal structure of external inputs. These emergent properties of information representation imply that the brain may adopt a dual coding system. Recent experimental data suggest that internally driven and externally driven dynamics coexist and work together in the cortex.

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126. Neurons with Two Sites of Synaptic Integration Learn Invariant Representations

Konrad P. Körding, Peter König

https://doi.org/10.1162/089976601317098547 | pp. 2823-2849

Neurons in mammalian cerebral cortex combine specific responses with respect to some stimulus features with invariant responses to other stimulus features. For example, in primary visual cortex, complex cells code for orientation of a contour but ignore its position to a certain degree. In higher areas, such as the inferotemporal cortex, translation-invariant, rotation-invariant, and even view point-invariant responses can be observed. Such properties are of obvious interest to artificial systems performing tasks like pattern recognition. It remains to be resolved how such response properties develop in biological systems. Here we present an unsupervised learning rule that addresses this problem. It is based on a neuron model with two sites of synaptic integration, allowing qualitatively different effects of input to basal and apical dendritic trees, respectively. Without supervision, the system learns to extract invariance properties using temporal or spatial continuity of stimuli. Furthermore, top-down information can be smoothly integrated in the same framework. Thus, this model lends a physiological implementation to approaches of unsupervised learning of invariant-response properties.

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127. Linear Constraints on Weight Representation for Generalized Learning of Multilayer Networks

Masaki Ishii, Itsuo Kumazawa

https://doi.org/10.1162/089976601317098556 | pp. 2851-2863

In this article, we present a technique to improve the generalization ability of multilayer neural networks. The proposed method introduces linear constraints on weight representation based on the invariance natures of training targets. We propose a learning method that introduces effective linear constraints into an error function as a penalty term. Furthermore, introduction of such constraints leads to reduction of the VC dimension of neural networks. We show bounds on the VC dimension of the neural networks with such constraints. Finally, we demonstrate the effectiveness of the proposed method by some experiments.

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128. Feedforward Neural Network Construction Using Cross Validation

Rudy Setiono

https://doi.org/10.1162/089976601317098565 | pp. 2865-2877

This article presents an algorithm that constructs feedforward neural networks with a single hidden layer for pattern classification. The algorithm starts with a small number of hidden units in the network and adds more hidden units as needed to improve the network's predictive accuracy. To determine when to stop adding new hidden units, the algorithm makes use of a subset of the available training samples for cross validation. New hidden units are added to the network only if they improve the classification accuracy of the network on the training samples and on the cross-validation samples. Extensive experimental results show that the algorithm is effective in obtaining networks with predictive accuracy rates that are better than those obtained by state-of-the-art decision tree methods.

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129. Index

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